Optimal Passive LC Compensation Under Nonsinusoidal Voltage: Extended Version

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***Abstract*—Passive compensation of nonsinusoidal circuits requires connecting source-current minimization to realizable networks of positive inductors and capacitors. This paper presents a time-domain design and synthesis method for lossless shunt networks using waveform inner products, without Fourier decomposition. Under fixed nonzero periodic terminal voltage with zero mean, finite cyclic closure, and forced periodic response, a geometric projection determines the unique optimal compensating current. When the compensable current space is two-dimensional, a rational closed form determines both elements of a series inductor-capacitor branch, with necessary and sufficient conditions for positive finite values to attain the global minimum of source root-mean-square current. When these conditions fail, boundary limits determine the infimum. Explicit examples with three and four directions exclude universal rational and radical formulas for optimal single-branch design, respectively. A constructive positivity criterion yields Cauer and Foster realizations with at most two elements per reactive direction when network order is unrestricted. In a 230-V comparison, equally optimal networks differ by approximately eightfold in mean stored energy. Simulink studies of thyristor-switched banks assess losses, source impedance, and switching limits beyond the ideal design assumptions.**

***Index Terms*—Passive compensation, LC networks, network synthesis, nonsinusoidal circuits, passive realizability, time-domain design, current minimization, thyristor-switched banks.**

I. Introduction

A LOSSLESS two-terminal network of capacitors and inductors, connected in parallel with a load, changes the source current without changing the load mean power when the terminal voltage is fixed and both networks are in periodic steady state. Under a periodic non-sinusoidal voltage, the design of such a network for minimum root-mean-square (RMS) source current raises three questions that become nontrivial beyond the sinusoidal case: which currents a lossless network can draw at a given voltage waveform, which element values produce the best of them, and whether those values admit a closed formula. Compensators of this kind are built in fixed form or in thyristor-switched steps [1]–[3], and their element values and switching decisions are computed by a controller from sampled voltage and current. This paper answers the three questions in the time domain, using temporal inner products and, where necessary, algebraic root isolation. The design requires no Fourier decomposition; spectral calculations are reserved for comparisons and independent checks.

The time-domain treatment of single elements is classical. Fryze's active current, the component of the load current with the waveform of the voltage, is the projection that the Fryze–Buchholz–Depenbrock method extends to multi-wire systems [4]. Kusters and Moore [5] resolved the load current into the components with the waveforms of the voltage, of its derivative and of its integral, each coefficient being a mean of products over one period, and showed that the optimal shunt capacitor or reactor is the one that cancels the corresponding component when its coefficient is negative; their statement that no passive linear compensation exists when both coefficients are positive was restricted, in the closure to Willems' discussion of the paper, to shunt capacitors and reactors alone. Page [6] determined both coefficients simultaneously from a two-by-two system of inner products and noted that both can come out negative, in which case the best compensator is a capacitor in parallel with an inductor; for an inductive load the same system returns a negative inductance [7]. A single capacitor does not in general reduce the source current to the active current under a distorted voltage [8], [9]. Lev-Ari and Stanković [10] placed these computations in a Hilbert space of periodic waveforms, where compensation onto specified current subspaces is expressed through orthogonal projections and the optimal capacitor per phase is the projection of the current onto the voltage derivative when the angle between them is obtuse; improvements beyond one element per phase are referred there to switching converters. The decomposition used in this paper is the power-free one of [11], extended in [12]: the load current is split, by projection and rejection with respect to the voltage, into an active current, a quadrature current that passive elements can cancel, and a remainder, with no power quantity involved; in the harmonic language of the currents' physical components theory these correspond to active and reactive currents, with the remainder also including current components absent from the voltage spectrum when present [13], [14].

Networks with more than one element have been designed in the frequency domain. Czarnecki [15] proved that a reactance one-port whose susceptance equals the negative of the load susceptance at every harmonic cancels the reactive current completely and needs no more than twice as many elements as harmonics. In [16] he showed that such a one-port has twice as many parameters as interpolation conditions, so that half

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of them are free, and that a single series LC branch with both elements free has, for an inductive load with a dominant fundamental, no minimum of reactive power, its infimum being approached as the inductance grows without bound; fixing one element is one way to define an attained design problem, which [7] does in the time domain by an iteration on measured inner products, for a capacitor bank switched in binary steps. Jeltsema, van der Woude and Hartman [17] rewrote the same decomposition as projections onto the harmonic components of the voltage and their derivatives and, for a load fed by two harmonics, obtained a three-element compensator by solving the differential equation of a series branch for a prescribed current. The realization of a prescribed lossless immittance is classical [18]–[20]; the number of elements and the structure of a passive realization are not fixed by the immittance alone [21], [22], and [1] collects the Foster and Cauer forms of reactance compensators.

When the voltage is distorted and the load non-linear, the single LC branch with both elements free has been designed by direct search [23], by penalty-function optimization [24], by minimizing voltage distortion subject to power-factor, resonance and commercial-value constraints [25] and, with line parameters estimated on line, in [26]; banks of resonant branches have been designed by fixing the resonant frequencies in advance and solving a linear system for the element values [3], with a linear program that keeps every element positive [27], and by a genetic algorithm over a catalogue of topologies [28]. These numerical and fixed-pole designs do not establish a universal closed form for the globally optimal single series branch with both elements free. Geometric algebra has been used to represent power and impedance under non-sinusoidal conditions [29]–[31], and later for compensation theorems valid at all harmonics at once [32] and for a time-domain framework of multiphase circuits [33]; the authors' earlier passive compensation with geometric algebra [34] was formulated in the frequency domain.

The cited approaches do not jointly provide a temporal characterization of all compensable currents, a classification of universal formulas for a series LC branch with both elements free, and a waveform-based criterion for that branch to attain the full compensable-current optimum. The last question extends the sign conditions for parallel elements in [5], [6], [10]. The design problem is to connect the admissible currents, positive realizability and element recovery through the subspace generated by the voltage.

This paper supplies the three. The voltage waveform generates, under the operator $-d^2/dt^2$, a subspace whose dimension m, the reactive dimension of the voltage, is half the smallest differential order of an even-derivative relation satisfied by the voltage; the derivative of that subspace is shown to be exactly the set of currents that a lossless network can draw, and the optimal compensating current is the negative of the orthogonal projection of the load current onto it. The construction is written in geometric algebra, where a blade represents a subspace and a contraction followed by blade inversion gives a projection and the compatibility of a current with a network is the vanishing of an outer product; each statement is accompanied by its matrix form. The contributions are the following.

1) A time-domain description of the compensable currents, computed from the voltage waveform, with the unique optimal projection. The capacitor, reactor and parallel LC designs of [5], [6] are recovered as projections onto the subspaces spanned by the derivative and the integral of the voltage, restricted to positive elements.
2) For $m = 2$, a rational closed form for the series LC branch with both elements free, with the criterion that a positive finite branch drawing the optimal current exists if and only if three nonzero quantities computed from the waveforms have the same sign; when it fails, no positive finite two-element branch attains the infimum. The example of [17] illustrates this boundary behavior; the three-direction example of [16] is audited separately. For explicit examples with $m = 3$ and $m = 4$, Galois-group computations show that the optimum is a root of an irreducible polynomial with Galois group S_4, which excludes a universal rational formula, and S_7, which excludes a universal formula by radicals; the element values are then roots of a polynomial in the inner products of the waveforms, a procedure distinct from the constrained searches of [23]–[26].
3) A time-domain synthesis of the free-order compensator, with a Bezoutian criterion of positive realness and Cauer or Foster extraction, that yields a one-parameter family of positive realizations, a subfamily within the realization freedom counted in [16]. A separate free-pole three-element family recovers the network of [17] within its published rounding. The geometry extends to banks, where branches of fixed shape with non-negative weights form a convex cone treated by Karush–Kuhn–Tucker conditions in reciprocal frames, thyristor-switched branches form a finite set of states compared by their costs, and the state of each branch, written in the idempotent–nilpotent basis of the two-dimensional geometric algebra, gives an energy bound to admit or reject a connection.
4) A quantitative comparison with the published designs of [15]–[17] and with capacitor, fixed-inductance, direct-search, fixed-pole and linear-programming designs on common test cases, reporting source current, stored energy and element count; and a closed-loop Simulink study of a bank with a fixed capacitor and two thyristor-switched series LC branches, with idealized valves, branch losses, source impedance and voltage and current limits, the load being represented by its current waveform. No hardware prototype is reported, and execution on an embedded controller is not evaluated; host-side timing is reported separately.

The rest of the paper is organized as follows. Section II states the hypotheses, in particular the fixed terminal voltage, the finite closure of the voltage as a model hypothesis and the forced periodic response of the compensator, and proves the projection theorem. Section III treats the small networks, the formula for $m = 2$ and the algebraic limits for $m \geq 3$. Section IV gives the synthesis and the banks. Sections V

and VI report the comparison and the simulation study, and Section VII discusses the limitations.

II. Geometric Framework

A. Waveforms, Operators and Hypotheses

Let $\mathcal{H}$ be the space of real periodic waveforms of period T with finite mean square. A waveform $x(t)$ is a vector $\boldsymbol{x}$ of $\mathcal{H}$ and is written in bold lowercase; multivectors of the algebra built on $\mathcal{H}$ are written in bold capitals, and scalars, including norms, in plain lowercase. The inner product and the norm are

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \frac{1}{T} \int_0^T x(t)\, y(t)\, dt, \qquad x = \|\boldsymbol{x}\| = \langle \boldsymbol{x}, \boldsymbol{x} \rangle^{1/2}, \quad (1)$$

so that x is the RMS value of the waveform. On the zero-mean subspace $\mathcal{H}_0$, with the periodic Sobolev domains H^1 for D and H^2 for A, the derivative $\mathsf{D} = d/dt$ satisfies $\langle \mathsf{D}\boldsymbol{x}, \boldsymbol{y} \rangle = -\langle \boldsymbol{x}, \mathsf{D}\boldsymbol{y} \rangle$, by integration by parts over one period, that is, $\mathsf{D}^* = -\mathsf{D}$; the zero-mean periodic antiderivative $\mathsf{J} = \mathsf{D}^{-1}$ is defined on $\mathcal{H}_0$; and $\mathsf{A} = -\mathsf{D}^2 = \mathsf{D}^*\mathsf{D}$ is self-adjoint and positive definite, since $\langle \mathsf{A}\boldsymbol{x}, \boldsymbol{x} \rangle = \|\mathsf{D}\boldsymbol{x}\|^2 > 0$ for $\boldsymbol{x} \neq \boldsymbol{0}$. For $\boldsymbol{x}$ in the corresponding operator domains:

$$\langle \boldsymbol{x}, \mathsf{D}\boldsymbol{x} \rangle = 0, \qquad \langle \boldsymbol{x}, \mathsf{J}\boldsymbol{x} \rangle = 0, \qquad \langle \mathsf{J}\boldsymbol{x}, \mathsf{D}\boldsymbol{x} \rangle = -x^2. \quad (2)$$

The first is $\mathsf{D}^* = -\mathsf{D}$ applied to $\langle \boldsymbol{x}, \mathsf{D}\boldsymbol{x} \rangle$; the second is the first applied to $\mathsf{J}\boldsymbol{x}$; the third is $\langle \mathsf{J}\boldsymbol{x}, \mathsf{D}\boldsymbol{x} \rangle = -\langle \mathsf{D}\mathsf{J}\boldsymbol{x}, \boldsymbol{x} \rangle$.

In the computations a period is represented by an odd number N of equally spaced samples, the inner product by $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{y}/N$, and D by the skew-symmetric differentiation matrix of the periodic trigonometric interpolant of the samples, a circulant matrix computed once in closed form. The interpolant has degree at most $(N-1)/2$; the differentiation matrix then has only the constant vector in its kernel, and the following identities hold on its zero-mean subspace. The equality between the sampled and the continuous RMS values is exact for the interpolant, not for an arbitrary continuous signal through the same samples.

The hypotheses are the following.

- (H1) The nonzero terminal voltage $\boldsymbol{u} \in \mathcal{H}_0$ is fixed: the compensator does not change it. This is exact for a source of zero internal impedance and is relaxed in Section VI.
- (H2) The load is in periodic steady state; its current $\boldsymbol{i} \in \mathcal{H}$ may contain components outside the span of $\boldsymbol{u}$ and its derivatives, as with a rectifier.
- (H3) (Finite closure.) There is a monic polynomial μ_u of minimal degree m such that $\mu_u(\mathsf{A})\boldsymbol{u} = \boldsymbol{0}$; m is the reactive dimension of $\boldsymbol{u}$. This is an exact model hypothesis. On a sampled record its numerical consistency is assessed through rank and the relative residual of the least-squares expression of $\mathsf{A}^m\boldsymbol{u}$ in terms of $\boldsymbol{u}, \mathsf{A}\boldsymbol{u}, \dots, \mathsf{A}^{m-1}\boldsymbol{u}$; it is not implied by periodicity.
- (H4) The compensator is a finite interconnection of positive capacitors and inductors, and only its forced periodic response is considered. Its admittance has no pole at a frequency of the voltage; in time-domain form, the denominator of the admittance, evaluated at $-\mathsf{A}$, is invertible on the subspace $\mathcal{E}_u$ defined below. The free oscillations excited at connection, which an ideal network does not damp, are treated in Section VI.

The source current is $\boldsymbol{i}_s = \boldsymbol{i} + \boldsymbol{q}$, with $\boldsymbol{q}$ the compensator current, and the objective is $\|\boldsymbol{i} + \boldsymbol{q}\|^2$, which fixes the conduction losses in a resistance in series with the source.

B. Reactive Dimension and Compensable Subspace

Definition 1. The cyclic subspace of the voltage and the compensable subspace are

$$\mathcal{E}_u = \operatorname{span}\{\boldsymbol{u}, \mathsf{A}\boldsymbol{u}, \mathsf{A}^2\boldsymbol{u}, \dots\}, \qquad \mathcal{Q}_u = \mathsf{D}\,\mathcal{E}_u. \quad (3)$$

Under (H3), $\dim \mathcal{E}_u = m$ and $\{\boldsymbol{u}, \mathsf{A}\boldsymbol{u}, \dots, \mathsf{A}^{m-1}\boldsymbol{u}\}$ is a basis of $\mathcal{E}_u$.

Lemma 1. *Under (H3): (a) μ_u has m distinct positive roots; (b) $\mathcal{E}_u \perp \mathcal{Q}_u$, in particular $\boldsymbol{u} \perp \mathcal{Q}_u$; (c) $\mathsf{J}\boldsymbol{u} \in \mathcal{Q}_u$, and $\mathcal{Q}_u = \operatorname{span}\{\mathsf{D}\boldsymbol{u}, \mathsf{D}^3\boldsymbol{u}, \dots, \mathsf{D}^{2m-1}\boldsymbol{u}\}$ has dimension m.*

Part (a) is the spectral theorem for the restriction of A to $\mathcal{E}_u$, a cyclic self-adjoint positive operator; (b) follows by moving derivatives across the inner product with (2); (c) from $\mu_u(\mathsf{A})\boldsymbol{u} = \boldsymbol{0}$ with a non-zero constant term (Section S-I of the Supplementary Material).

The projection and free-order synthesis do not require the roots of μ_u; their construction uses its coefficients of μ_u, obtained from the first linear dependence among $\boldsymbol{u}, \mathsf{A}\boldsymbol{u}, \dots$ in the time domain. For $m = 2$, $\mu_u(\lambda) = \lambda^2 - s\lambda + p$, that is,

$$\mathsf{D}^4\boldsymbol{u} + s\,\mathsf{D}^2\boldsymbol{u} + p\,\boldsymbol{u} = \boldsymbol{0}, \qquad s > 0, \quad p > 0, \quad s^2 > 4p, \quad (4)$$

where the sign conditions follow from (a).

Remark 1 (Frequency viewpoint). If $\boldsymbol{u}$ is a finite sum of sinusoids with distinct angular frequencies $\omega_1, \dots, \omega_m$, then A multiplies the k-th sinusoid by ω_k^2, $\mathcal{E}_u$ is the span of the m sinusoids, $\mu_u(\lambda) = \prod_k (\lambda - \omega_k^2)$, and $\mathcal{Q}_u$ is spanned by the m sinusoids in quadrature. The paper uses this correspondence only to compare with published designs.

Proposition 1. *Under (H1)–(H4), the current $\boldsymbol{q}$ drawn by a lossless network of admittance Y at the voltage $\boldsymbol{u}$ belongs to $\mathcal{Q}_u$.*

Proof: The admittance of a finite network of positive capacitors and inductors is a reactance function, in particular an odd rational function [18], so $Y(s) = s\,F(s^2)$ with F rational, and the forced periodic response is $\boldsymbol{q} = Y(\mathsf{D})\boldsymbol{u} = \mathsf{D}\,F(-\mathsf{A})\boldsymbol{u}$. By (H4) the denominator of $F(-\mathsf{A})$ is invertible on $\mathcal{E}_u$; the inverse of an invertible polynomial in A on the finite-dimensional invariant subspace $\mathcal{E}_u$ is itself a polynomial in A, so $F(-\mathsf{A})\boldsymbol{u} \in \mathcal{E}_u$ and $\boldsymbol{q} \in \mathsf{D}\,\mathcal{E}_u$. ∎

The converse is Theorem 5 of Section IV: every element of $\mathcal{Q}_u$ is the current of some network of positive elements. Hence $\mathcal{Q}_u$, which depends on the voltage alone, is the set of compensable currents. Three conditions on a current $\boldsymbol{q}$ must be kept apart: $\langle \boldsymbol{q}, \boldsymbol{u} \rangle = 0$, which every element of $\mathcal{Q}_u$ satisfies by Lemma 1(b) but which a negative capacitance also satisfies; membership in $\mathcal{Q}_u$; and realizability by a prescribed network with positive elements, which is the question of Section III.

C. The Projection Theorem

Theorem 1. *Let $\boldsymbol{q}^* = -\operatorname{proj}_{\mathcal{Q}_u} \boldsymbol{i}$. For every $\boldsymbol{q} \in \mathcal{Q}_u$,*

$$\|\boldsymbol{i}+\boldsymbol{q}\|^2 = \|\boldsymbol{i}+\boldsymbol{q}^*\|^2 + \|\boldsymbol{q}-\boldsymbol{q}^*\|^2, \tag{5}$$

so $\boldsymbol{q}^*$ is the unique minimizer of the RMS source current over $\mathcal{Q}_u$.

Proof: $\boldsymbol{i}+\boldsymbol{q} = (\boldsymbol{i}-\operatorname{proj}_{\mathcal{Q}_u}\boldsymbol{i}) + (\boldsymbol{q}+\operatorname{proj}_{\mathcal{Q}_u}\boldsymbol{i})$; the first term is orthogonal to $\mathcal{Q}_u$ and the second belongs to it. ∎

Corollary 1. *Let $\boldsymbol{i}_a = \langle \boldsymbol{i}, \boldsymbol{u}\rangle\, \boldsymbol{u}/u^2$, $\boldsymbol{i}_q = \operatorname{proj}_{\mathcal{Q}_u}\boldsymbol{i}$ and $\boldsymbol{i}_r = \boldsymbol{i}-\boldsymbol{i}_a-\boldsymbol{i}_q$. The three components are mutually orthogonal, $\boldsymbol{i}+\boldsymbol{q}^* = \boldsymbol{i}_a+\boldsymbol{i}_r$, and*

$$\|\boldsymbol{i}+\boldsymbol{q}^*\|^2 = i_a^2 + i_r^2. \tag{6}$$

For $i_a > 0$, the current efficiency index $\lambda = i_a/i_s$, the ratio between the RMS values of the active and the source currents, is therefore bounded for every lossless compensator by $\lambda_{\max} = i_a/(i_a^2+i_r^2)^{1/2}$, reached by $\boldsymbol{q}^*$, and $\lambda_{\max} = 1$ if and only if $\boldsymbol{i}_r = \boldsymbol{0}$, that is, if and only if $\boldsymbol{i} \in \operatorname{span}\{\boldsymbol{u}\} \oplus \mathcal{Q}_u$.

Proof: $\boldsymbol{i}_a \perp \boldsymbol{i}_q$ because $\boldsymbol{u} \perp \mathcal{Q}_u$; $\boldsymbol{i}_r$ is orthogonal to both by the definition of the projections. ∎

In the language of [11], $\boldsymbol{i}_a$ is the projection $(\boldsymbol{i}\cdot\boldsymbol{u})\boldsymbol{u}^{-1}$ and $\boldsymbol{i}_q$ is the projection onto $\mathcal{Q}_u$ of the rejection $(\boldsymbol{i}\wedge\boldsymbol{u})\boldsymbol{u}^{-1}$, which equals $\operatorname{proj}_{\mathcal{Q}_u}\boldsymbol{i}$ because $\boldsymbol{u} \perp \mathcal{Q}_u$. In the frequency viewpoint of Remark 1, $\boldsymbol{i}_q$ is the reactive current of the CPC theory, $\boldsymbol{i}_r$ collects the scattered current and the components of $\boldsymbol{i}$ at frequencies absent from $\boldsymbol{u}$, and $\lambda_{\max}$ is, in the terms of [14], the maximal power factor of a shunt reactance compensator. Here the three components and the bound follow from the orthogonality of two subspaces defined by the voltage waveform.

The projection is computed with the basis waveforms $\boldsymbol{b}_j = \mathsf{D}\mathsf{A}^{j-1}\boldsymbol{u}$, $j = 1,\dots,m$, of $\mathcal{Q}_u$: with $G_{jk} = \langle \boldsymbol{b}_j, \boldsymbol{b}_k\rangle$ and $\beta_j = \langle \boldsymbol{b}_j, \boldsymbol{i}\rangle$,

$$c = -G^{-1}\beta, \qquad \boldsymbol{q}^* = \sum_{j=1}^{m} c_j \boldsymbol{b}_j = \mathsf{D}\, h(\mathsf{A})\boldsymbol{u}, \tag{7}$$

with $h(\lambda) = \sum_{j=1}^{m} c_j \lambda^{j-1}$, where the linear system is solved by orthogonal factorization rather than by an explicit inverse. The polynomial h, of degree less than m, is the datum passed to the synthesis of Section IV. The basis waveforms are obtained by repeated differentiation of $\boldsymbol{u}$ or, in a controller, by the two-state resonant filters of Section VI; G and β require $m(m+3)/2$ inner products of N samples.

D. Geometric-Algebra Form

The geometric algebra of waveforms is generated by the sampled waveform space with the inner product (1): waveforms are vectors, $\boldsymbol{x}^2 = x^2$, and the geometric product is $\boldsymbol{x}\boldsymbol{y} = \boldsymbol{x}\cdot\boldsymbol{y} + \boldsymbol{x}\wedge\boldsymbol{y}$. A subspace is represented by a blade; for $\mathcal{Q}_u$,

$$\boldsymbol{B} = \boldsymbol{b}_1 \wedge \boldsymbol{b}_2 \wedge \cdots \wedge \boldsymbol{b}_m, \tag{8}$$

which changes by a scalar factor when the basis changes. With $\widetilde{\boldsymbol{B}}$ the reverse of $\boldsymbol{B}$, the scalar $\langle \widetilde{\boldsymbol{B}}\boldsymbol{B}\rangle_0 = \|\boldsymbol{B}\|^2$ is the Gram determinant of the basis, positive for independent $\boldsymbol{b}_j$, whereas $\boldsymbol{B}^2$ may be negative; the inverse of the simple blade is $\boldsymbol{B}^{-1} = \widetilde{\boldsymbol{B}}/\|\boldsymbol{B}\|^2$. The projection onto the subspace of $\boldsymbol{B}$ is $(\boldsymbol{x} \lrcorner \boldsymbol{B})\boldsymbol{B}^{-1}$, with $\lrcorner$ the left contraction, so Theorem 1 reads

$$\boldsymbol{q}^* = -(\boldsymbol{i} \lrcorner \boldsymbol{B})\, \boldsymbol{B}^{-1}. \tag{9}$$

The reciprocal frame $\boldsymbol{b}^j = (-1)^{j-1}(\boldsymbol{b}_1 \wedge \cdots \wedge \check{\boldsymbol{b}}_j \wedge \cdots \wedge \boldsymbol{b}_m)\boldsymbol{B}^{-1}$, where $\check{\boldsymbol{b}}_j$ denotes omission, satisfies $\boldsymbol{b}_j \cdot \boldsymbol{b}^k = \delta_{jk}$, and the coefficients of (7) are $c_j = -\langle \boldsymbol{i}, \boldsymbol{b}^j\rangle$: the reciprocal frame is the geometric form of the inverse Gram matrix. Only inner products of waveforms enter these expressions; no algebra of dimension 2^N is materialized, and every formula has the matrix counterpart written beside it.

The geometric form also recovers element values when the circuit law is linear in the unknown parameters; for a series LC branch, $\boldsymbol{u} = L\,\mathsf{D}\boldsymbol{q} + (1/C)\,\mathsf{J}\boldsymbol{q}$. The following statement gives the parameters, and the condition for their existence, as quotients of blades.

Proposition 2. *Let $\boldsymbol{u} = \sum_{k=1}^{n} \theta_k T_k \boldsymbol{q}$ be the relation between the voltage and the current of a network with parameters $\theta_1,\dots,\theta_n$, where the T_k are known linear operators, and let $\boldsymbol{x}_k = T_k\boldsymbol{q}$ and $\boldsymbol{X} = \boldsymbol{x}_1 \wedge \cdots \wedge \boldsymbol{x}_n \neq \boldsymbol{0}$. Then a parameter vector satisfying the relation exists if and only if*

$$\boldsymbol{u} \wedge \boldsymbol{X} = \boldsymbol{0}, \tag{10}$$

and it is then unique and given by $\theta_k = \boldsymbol{X}_k\boldsymbol{X}^{-1}$, where $\boldsymbol{X}_k$ is $\boldsymbol{X}$ with $\boldsymbol{x}_k$ replaced by $\boldsymbol{u}$.

When the relation holds, multilinearity and antisymmetry of the outer product give $\boldsymbol{X}_k = \theta_k \boldsymbol{X}$; conversely, (10) with $\boldsymbol{X} \neq \boldsymbol{0}$ means that $\boldsymbol{u}$ lies in the span of the independent $\boldsymbol{x}_k$ (Section S-I of the Supplementary Material).

The matrix counterpart is Cramer's rule on the Gram system $\sum_k \theta_k \langle \boldsymbol{x}_j, \boldsymbol{x}_k\rangle = \langle \boldsymbol{x}_j, \boldsymbol{u}\rangle$, which has a solution satisfying the relation exactly if and only if (10) holds. When (10) fails, $\boldsymbol{X}_k\boldsymbol{X}^{-1}$ is not a scalar; its scalar part solves the least-squares version of the relation, which is a fit of the voltage equation and not the current-minimization problem of Theorem 1. Positivity of the θ_k is a further condition, not implied by (10): compatibility, like orthogonality to $\boldsymbol{u}$, does not guarantee passivity.

E. What the Framework Decides

The voltage fixes $\mathcal{Q}_u$; the voltage and load current together determine $\boldsymbol{q}^*$, $\boldsymbol{i}_r$ and the bound (6). It leaves open two questions that the next sections answer. Whether a prescribed network with few positive elements can draw $\boldsymbol{q}^*$ is the compatibility condition (10) together with positivity: Section III solves it for one and two elements, using the appropriate one- or two-direction subspace while retaining $\boldsymbol{q}^*$ for the optimum over the full $\mathcal{Q}_u$. For larger m, finite interior candidates are algebraic over the field generated by the data; Section III-D states the precise limits on universal formulas. Whether some network draws $\boldsymbol{q}^*$ when its order is free is answered affirmatively in Section IV, which also shows that the network is not unique, in agreement with the general results on passive realizations [21], [22].

III. Small Compensators: Closed Forms and Their Limits

This section treats networks with one or two elements. The one-element and the parallel designs are classical and are restated in the language of Section II, because the series branch is built from the same two coordinates. The series branch with both elements free is then solved: a rational closed form with a sign criterion when $m = 2$, and two loads with $m = 3$ and $m = 4$ for which no formula of a given type can exist.

A. One Element and Two Parallel Elements

A capacitor draws $C\,\mathsf{D}\boldsymbol{u}$ and an inductor draws $\Gamma\,\mathsf{J}\boldsymbol{u}$ with $\Gamma = 1/L$, so their admissible currents are the rays spanned by $\mathsf{D}\boldsymbol{u}$ and by $\mathsf{J}\boldsymbol{u}$ with non-negative scale. On a ray, $\|\boldsymbol{i} + C\,\mathsf{D}\boldsymbol{u}\|$ is smallest when $C\,\mathsf{D}\boldsymbol{u}$ is the projection of $-\boldsymbol{i}$ onto the line, whose coordinate is the inner product with the inverse vector $(\mathsf{D}\boldsymbol{u})^{-1} = \mathsf{D}\boldsymbol{u}/\|\mathsf{D}\boldsymbol{u}\|^2$; the positive part enforces the sign constraint:

$$\begin{aligned} C^* &= [C_0]_+, \qquad C_0 = -\boldsymbol{i}\cdot(\mathsf{D}\boldsymbol{u})^{-1}, \\ \Gamma^* &= [\Gamma_0]_+, \qquad \Gamma_0 = -\boldsymbol{i}\cdot(\mathsf{J}\boldsymbol{u})^{-1}, \end{aligned} \tag{11}$$

with $[x]_+ = \max(0, x)$; a zero value means that the element is omitted. These are the capacitor and the reactor of Kusters and Moore [5], and the rule of the obtuse angle of [10].

A capacitor in parallel with an inductor draws $C_\parallel\,\mathsf{D}\boldsymbol{u} + \Gamma_\parallel\,\mathsf{J}\boldsymbol{u}$. Let $\boldsymbol{B}_2 = \mathsf{D}\boldsymbol{u} \wedge \mathsf{J}\boldsymbol{u}$ be the blade of the plane spanned by the two waveforms; by (2), $\|\boldsymbol{B}_2\|^2 = \|\mathsf{D}\boldsymbol{u}\|^2\|\mathsf{J}\boldsymbol{u}\|^2 - u^4$, which is positive unless $m = 1$. The unconstrained pair draws the projection of $-\boldsymbol{i}$ onto the plane, and its values are the coordinates of that projection in the frame $\{\mathsf{D}\boldsymbol{u}, \mathsf{J}\boldsymbol{u}\}$. Because the two waveforms are not orthogonal, the coordinates are read with the reciprocal frame $\{\boldsymbol{b}^1, \boldsymbol{b}^2\} = \{\mathsf{J}\boldsymbol{u}\,\boldsymbol{B}_2^{-1},\ -\mathsf{D}\boldsymbol{u}\,\boldsymbol{B}_2^{-1}\}$, in which $\boldsymbol{b}^j$ is the vector of the plane orthogonal to the other basis vector, scaled so that $\boldsymbol{b}_j \cdot \boldsymbol{b}^j = 1$; since the $\boldsymbol{b}^j$ lie in the plane, the coordinates of the projection of $-\boldsymbol{i}$ are the inner products of $-\boldsymbol{i}$ itself with them:

$$\begin{aligned} \boldsymbol{q}_\parallel &= -(\boldsymbol{i} \lrcorner \boldsymbol{B}_2)\,\boldsymbol{B}_2^{-1} = C_\parallel\,\mathsf{D}\boldsymbol{u} + \Gamma_\parallel\,\mathsf{J}\boldsymbol{u}, \\ C_\parallel &= -\boldsymbol{i}\cdot\big(\mathsf{J}\boldsymbol{u}\,\boldsymbol{B}_2^{-1}\big), \qquad \Gamma_\parallel = \boldsymbol{i}\cdot\big(\mathsf{D}\boldsymbol{u}\,\boldsymbol{B}_2^{-1}\big). \end{aligned} \tag{12}$$

In matrix form, (12) is Cramer's rule on the Gram matrix of the pair, whose off-diagonal entry is $\langle \mathsf{D}\boldsymbol{u}, \mathsf{J}\boldsymbol{u}\rangle = -u^2$; it is the system of Page [6]. The pair is realizable when $C_\parallel \geq 0$ and $\Gamma_\parallel \geq 0$. Otherwise the minimum over the cone of positive pairs lies on one of its faces, capacitor alone, inductor alone or no element, and is found by comparing them with (11); it is not found by setting the negative coordinate to zero, because the two waveforms are not orthogonal. When $m = 2$, $\boldsymbol{B}_2$ represents $\mathcal{Q}_u$ by Lemma 1(c) and $\boldsymbol{q}_\parallel = \boldsymbol{q}^*$, so a realizable pair attains the bound (6); when $m \geq 3$ it is the projection onto a plane inside $\mathcal{Q}_u$. None of this is new: it is [5], [6], [10] in the present language, and it is kept because the signed values (12) are the input of the series branch.

B. The Series Branch as a Curve of Rays

A series branch of inductance L and capacitance C obeys $\boldsymbol{u} = L\,\mathsf{D}\boldsymbol{q} + S\,\mathsf{J}\boldsymbol{q}$ with $S = 1/C$. Applying D gives $\mathsf{D}\boldsymbol{u} = (S - L\mathsf{A})\boldsymbol{q}$, so the forced periodic current is

$$\boldsymbol{q} = \Gamma\,\boldsymbol{w}_z, \qquad \boldsymbol{w}_z = (z - \mathsf{A})^{-1}\mathsf{D}\boldsymbol{u}, \qquad z = \frac{1}{LC}, \tag{13}$$

where the inverse acts on $\mathcal{Q}_u$ (with the same spectrum as $\mathcal{E}_u$) and exists if and only if z is not a root of μ_u, which is hypothesis (H4). The tuning z fixes the shape of the current and Γ its scale. The admissible currents of a single branch, $\{\Gamma\boldsymbol{w}_z : \Gamma \geq 0,\ z > 0,\ \mu_u(z) \neq 0\}$, form a union of rays indexed by z; for $m \geq 2$, unlike the fixed-generator sets of Section III-A, this union is neither a subspace nor a convex cone, and this is why the series branch behaves differently from the parallel pair. For fixed z, the best scale and the reduction of the squared RMS source current that it produces, $\Delta(z) = i^2 - \|\boldsymbol{i} + \Gamma^*(z)\boldsymbol{w}_z\|^2$, are

$$\Gamma^*(z) = \frac{[-\boldsymbol{i}\cdot\boldsymbol{w}_z]_+}{\boldsymbol{w}_z^2}, \qquad \Delta(z) = \frac{[-\boldsymbol{i}\cdot\boldsymbol{w}_z]_+^2}{\boldsymbol{w}_z^2}. \tag{14}$$

At an interior optimum with $\Gamma^* > 0$, the residual $\boldsymbol{i} + \Gamma^*\boldsymbol{w}_z$ is orthogonal to the tangent plane $\boldsymbol{w}_z \wedge \partial_z\boldsymbol{w}_z$ of the family; eliminating Γ leaves

$$\boldsymbol{i}\cdot\big[(\boldsymbol{w}_z \wedge \partial_z\boldsymbol{w}_z)\,\boldsymbol{w}_z\big] = 0, \tag{15}$$

which, by the identity $(\boldsymbol{a}\wedge\boldsymbol{b})\boldsymbol{a} = \boldsymbol{a}(\boldsymbol{a}\cdot\boldsymbol{b}) - \boldsymbol{b}\,a^2$, is the component of $\boldsymbol{i}$ along the change of shape orthogonal to the direction already optimized. Equation (15) is a condition of stationarity; the global optimum is selected by comparing its solutions with the boundaries of the admissible set.

For $m = 1$, write $\boldsymbol{q}^* = a\mathsf{D}\boldsymbol{u}$ and let $\lambda > 0$ be the root of μ_u. If $a > 0$, a capacitor $C = a$ draws $\boldsymbol{q}^*$; if $a < 0$, an inductor $L = -1/(a\lambda)$ does so, since $\mathsf{J}\boldsymbol{u} = -\mathsf{D}\boldsymbol{u}/\lambda$. For $a \neq 0$, every $L > 0$ with $S = \lambda L + 1/a > 0$ gives a positive series realization. If $a = 0$, the optimal choice is to omit the branch. Thus the optimal current is unique, whereas the positive series pair is not. The following statement treats $m \geq 2$.

Proposition 3 (Finite algebraic reduction)**.** *Let $\mu_u(\xi) = \sum_{j=0}^{m}\pi_j\xi^j$ with $\pi_m = 1$, and let*

$$\boldsymbol{W}(z) = \sum_{j=1}^{m}\pi_j\sum_{k=0}^{j-1} z^{\,j-1-k}\mathsf{A}^k\mathsf{D}\boldsymbol{u}, \tag{16}$$

a polynomial in z with vector coefficients. Then $(z - \mathsf{A})\boldsymbol{W}(z) = \mu_u(z)\,\mathsf{D}\boldsymbol{u}$, so $\boldsymbol{w}_z = \boldsymbol{W}(z)/\mu_u(z)$; with the scalar polynomials $a(z) = \boldsymbol{i}\cdot\boldsymbol{W}(z)$ and $b(z) = \boldsymbol{W}(z)^2$, whose coefficients are inner products of the waveforms, $\Gamma^*(z) = [-\mu_u(z)a(z)]_+/b(z)$ and $\Delta(z) = a(z)^2/b(z)$ when $\mu_u(z)a(z) < 0$, zero otherwise. The interior stationary shapes are the roots of

$$F(z) = 2a'(z)\,b(z) - a(z)\,b'(z), \qquad \deg F \leq 3m - 5, \tag{17}$$

and the infimum is determined by comparing the positive roots of F with $\Gamma^* > 0$, the absence of branch, the limits $z \to 0$ and $z \to \infty$, which are the inductor and the capacitor alone, and the one-sided limits at the roots of μ_u.

The identity for $\boldsymbol{W}$ is a telescoping sum together with $\mu_u(\mathsf{A})\mathsf{D}\boldsymbol{u} = \mathbf{0}$, and $(a^2/b)' = aF/b^2$; on the active set $a \neq 0$, stationary points therefore satisfy $F = 0$, whose leading terms cancel (Section S-II of the Supplementary Material). Proposition 3 addresses the question of formula type raised in the Introduction for a single branch: the tuning of a finite interior optimum is a root of a polynomial of degree at most $3m - 5$, and its element values follow rationally from that tuning and the inner products. For $m = 2$ this degree is one, which is the closed form of the next subsection; for $m = 3$ and $m = 4$ it is four and seven, the degrees of the two examples of Section III-D.

C. Two Reactive Directions: the Closed Form

Theorem 2. *Let (H1)–(H4) hold with $m = 2$, so that $\boldsymbol{u}$ satisfies* (4) *and $\boldsymbol{u} \wedge \mathsf{D}^2\boldsymbol{u} \neq \mathbf{0}$. Let $C_\parallel$ and $\Gamma_\parallel$ be the coordinates* (12)*, so that $\boldsymbol{q}^* = C_\parallel\, \mathsf{D}\boldsymbol{u} + \Gamma_\parallel\, \mathsf{J}\boldsymbol{u}$, and let*

$$K = p\, C_\parallel^2 - s\, C_\parallel \Gamma_\parallel + \Gamma_\parallel^2. \tag{18}$$

(i) *If $C_\parallel$, $\Gamma_\parallel$ and K are non-zero and of the same sign, the series branch*

$$L = \frac{\Gamma_\parallel}{K}, \qquad C = \frac{K}{p\, C_\parallel} \tag{19}$$

is positive and finite, draws $\boldsymbol{q}^*$, is the only series branch that draws it, and attains the bound (6) *over all lossless compensators.*

(ii) *Otherwise no positive finite series branch draws $\boldsymbol{q}^*$, no positive finite series branch is a local minimizer of the source current, and the infimum over positive finite branches is approached on the boundary of the admissible set: an element tending to zero or to infinity, which leaves the capacitor or the inductor alone, or z tending to a root of μ_u, where $\Gamma^* \to 0$.*

Proof: By Lemma 1(c), $\boldsymbol{B}_2$ represents $\mathcal{Q}_u$, so $\boldsymbol{q}^* = C_\parallel \mathsf{D}\boldsymbol{u} + \Gamma_\parallel \mathsf{J}\boldsymbol{u}$. Applying J^2 to (4) gives $\mathsf{J}^2\boldsymbol{u} = -(\mathsf{D}^2\boldsymbol{u} + s\boldsymbol{u})/p$, hence

$$\begin{aligned} \boldsymbol{x} &= \mathsf{D}\boldsymbol{q}^* = \Gamma_\parallel\, \boldsymbol{u} + C_\parallel\, \mathsf{D}^2\boldsymbol{u}, \\ \boldsymbol{y} &= \mathsf{J}\boldsymbol{q}^* = \Big(C_\parallel - \frac{s\Gamma_\parallel}{p}\Big)\boldsymbol{u} - \frac{\Gamma_\parallel}{p}\, \mathsf{D}^2\boldsymbol{u}, \end{aligned} \tag{20}$$

and

$$\boldsymbol{x} \wedge \boldsymbol{y} = -\frac{K}{p}\, \boldsymbol{u} \wedge \mathsf{D}^2\boldsymbol{u}. \tag{21}$$

(i) $K \neq 0$ gives $\boldsymbol{x} \wedge \boldsymbol{y} \neq \mathbf{0}$, and (10) holds because $\boldsymbol{u}$, $\boldsymbol{x}$ and $\boldsymbol{y}$ lie in the plane $\mathcal{E}_u = \operatorname{span}\{\boldsymbol{u}, \mathsf{D}^2\boldsymbol{u}\}$. Proposition 2 then gives a unique pair,

$$\begin{aligned} L &= (\boldsymbol{u} \wedge \mathsf{J}\boldsymbol{q}^*)(\mathsf{D}\boldsymbol{q}^* \wedge \mathsf{J}\boldsymbol{q}^*)^{-1}, \\ C &= (\mathsf{D}\boldsymbol{q}^* \wedge \mathsf{J}\boldsymbol{q}^*)(\mathsf{D}\boldsymbol{q}^* \wedge \boldsymbol{u})^{-1}, \end{aligned} \tag{22}$$

and expanding the blades with (20) gives (19). Since $p > 0$, L and C are positive and finite exactly when the three quantities are non-zero and of the same sign; the branch draws $\boldsymbol{q}^*$, the projection onto the whole $\mathcal{Q}_u$, so Theorem 1 gives the last claim. The vanishing of K is the vanishing of the blade (21).

(ii) If $\boldsymbol{q}^* = \mathbf{0}$, or one coordinate vanishes, or $K = 0$, the branch degenerates into no element, one element, or a current with a single reactive direction that no finite branch draws (Section S-III of the Supplementary Material). If the three quantities are non-zero and not of the same sign, a positive finite branch at an interior local minimum would have its residual orthogonal to $\boldsymbol{w}_z$ and $\partial_z \boldsymbol{w}_z$, which are independent for $m = 2$ and span $\mathcal{Q}_u$; it would then draw $\boldsymbol{q}^*$, which is impossible. Hence the infimum is approached on the boundary of the admissible set, as stated (Section S-III of the Supplementary Material). ∎

The coefficients s and p of (4) are the coordinates of $\mathsf{A}^2\boldsymbol{u} = s\,\mathsf{A}\boldsymbol{u} - p\,\boldsymbol{u}$ in the frame $\{\mathsf{A}\boldsymbol{u}, \boldsymbol{u}\}$ of $\mathcal{E}_u$: with $\boldsymbol{E}_2 = \mathsf{A}\boldsymbol{u} \wedge \boldsymbol{u}$,

$$s = \mathsf{A}^2\boldsymbol{u} \cdot \big(\boldsymbol{u}\, \boldsymbol{E}_2^{-1}\big), \qquad p = \mathsf{A}^2\boldsymbol{u} \cdot \big(\mathsf{A}\boldsymbol{u}\, \boldsymbol{E}_2^{-1}\big). \tag{23}$$

Fitting these two coordinates does not prove that a voltage has $m = 2$; hypothesis (H3) is checked by the residual of the linear dependence. The whole design of the branch is therefore the inner products that enter (12) and (23), the scalar K and the two quotients (19); no root is extracted.

Example. In normalized time $\tau = \omega_* t$ with $\omega_* = 2\pi \cdot 50$ rad/s, take the voltage given by the five samples per period $(1,\ 0.2,\ -0.7,\ -0.6,\ 0.1)$ V, whose periodic interpolant satisfies (4) with $s = 5$ and $p = 4$, and the load $\boldsymbol{i} = 0.8\,\boldsymbol{u} + 0.3\,\mathsf{D}\boldsymbol{u} + 0.7\,\mathsf{J}\boldsymbol{u}$, that is, a conductance, a capacitor and an inductor in parallel, all positive. The projection gives $C_\parallel = -0.3$ and $\Gamma_\parallel = -0.7$: the parallel pair would need negative elements, and the capacitor alone of (11) reduces the source current from 0.550 A to 0.505 A only. Here $K = -0.2$, so the three quantities are negative and Theorem 2(i) applies: $L = 3.5$ and $C = 1/6$ in normalized units, that is, $L = 11.14$ mH and $C = 530.5$ μF after dividing by ω_*. The branch draws $\boldsymbol{q}^* = -0.3\,\mathsf{D}\boldsymbol{u} - 0.7\,\mathsf{J}\boldsymbol{u}$ exactly, the source current becomes $0.8\,\boldsymbol{u}$, with RMS value 0.493 A, and the current efficiency index rises from 0.896 to one because $\boldsymbol{i}_r = \mathbf{0}$. Fig. 1(a) shows the source current of a single branch as a function of its tuning z, with the scale chosen by (14): it reaches the bound (6) at $z = 1/(LC) = 12/7$ and nowhere else. For this closure $K = (\Gamma_\parallel - C_\parallel)(\Gamma_\parallel - 4C_\parallel)$, so Theorem 2(i) admits the pairs with $C_\parallel, \Gamma_\parallel > 0$ outside the sector between the lines $\Gamma_\parallel = C_\parallel$ and $\Gamma_\parallel = 4C_\parallel$, and the pairs with $C_\parallel, \Gamma_\parallel < 0$ inside it, where the example lies; negative values of the parallel pair produce a positive series branch. On the axes, a positive single element can realize $\boldsymbol{q}^*$ when its surviving coefficient is positive. On $K = 0$ with $\boldsymbol{q}^* \neq 0$, the target has only one reactive direction and no positive finite series branch realizes it.

Discordant signs. When $m = 2$ and $C_\parallel > 0 > \Gamma_\parallel$, Theorem 2(ii) rules out a positive finite optimum. For the R–L example of [17], the largest boundary improvement occurs at the lower root of μ_u, with $\Gamma^* \to 0$, $L \to \infty$ and $C \to 0$; Fig. 1(a) marks this unattained infimum. This resembles the behavior in [16], whose three-direction example is checked independently in Section V. For other loads the relevant boundary must be selected by Proposition 3. For instance, if $\boldsymbol{i} = G\boldsymbol{u} - c\mathsf{D}\boldsymbol{u}$ with $G, c > 0$, the capacitor $C = c$ attains the bound, whereas a series branch approaches it only as $z \to \infty$, $\Gamma = cz$, $L \to 0$ and $C \to c$.

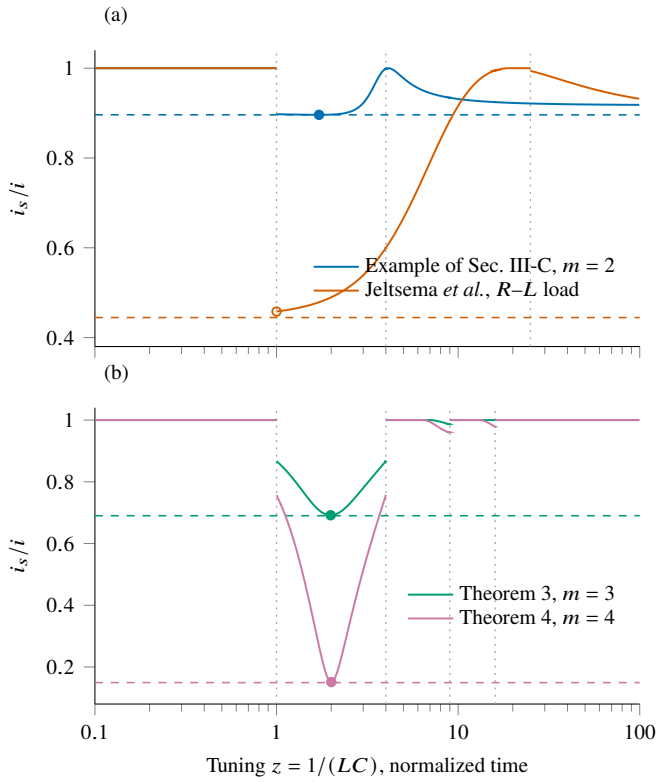


Fig. 1. RMS source current with a single series LC branch, relative to the uncompensated value, as a function of the tuning $z = 1/(LC)$ in normalized time, with the scale Γ chosen by (14) for each z. Dotted lines: roots of μ_u, where the branch resonates with the voltage and $\Gamma^* \to 0$; dashed lines: the bound (6) of each case. (a) $m = 2$: the example of Section III-C attains the bound at $z = 12/7$ (filled marker, Theorem 2(i)); the R–L load of [17] has discordant signs and its infimum (open marker) lies at the lower resonance and is not attained (Theorem 2(ii)). (b) The loads of Theorems 3 and 4, with their unique interior optima at $z = 1.9867$ and $z = 2.0024$.

D. Three and Four Reactive Directions: Algebraic Limits

For $m \geq 3$ a single branch draws the projection $\boldsymbol{q}^*$ onto the whole $\mathcal{Q}_u$ if and only if, with $\boldsymbol{x} = \mathsf{D}\boldsymbol{q}^*$ and $\boldsymbol{y} = \mathsf{J}\boldsymbol{q}^*$, $\boldsymbol{x} \wedge \boldsymbol{y} \neq \boldsymbol{0}$, the compatibility condition $\boldsymbol{u} \wedge \boldsymbol{x} \wedge \boldsymbol{y} = \boldsymbol{0}$ holds and the quotients (22) are positive; the compatibility condition, automatic for $m = 2$, fails in general because $\boldsymbol{u}$ has m directions and the branch law confines it to two. When it fails, no single branch attains the bound (6), and Proposition 3 must compare partial interior optima, when they exist, with boundary infima. Two loads, constructed in the time domain with rational data, show what kind of formula can and cannot give that optimum. In both, τ is normalized time of period 2π, $\mathsf{D} = d/d\tau$, and the load is $\boldsymbol{i} = G\boldsymbol{u} + h(\mathsf{A})\mathsf{D}\boldsymbol{u}$ with a conductance $G > 0$ that does not affect $\boldsymbol{q}^*$.

Theorem 3 ($m = 3$)**.** *Let* $u(\tau) = \sqrt{2}\,(4\cos^3\tau + 2\cos^2\tau - 2\cos\tau - 1)$*, for which* $\mu_u(\xi) = \xi^3 - 14\xi^2 + 49\xi - 36$*, and* $h(\xi) = (-13\xi^2 + 155\xi - 322)/180$*, so that* $h(\mathsf{A})\mathsf{D}\boldsymbol{u}$ *agrees on this excitation with the current of a capacitor* $11/20$*, an inductor* $10/17$ *and a series branch* $L = 4/3$*,* $C = 1/8$*, all positive and in parallel. The inner products among* $\mathsf{D}\boldsymbol{u}$*,* $\mathsf{AD}\boldsymbol{u}$*,* $\mathsf{A}^2\mathsf{D}\boldsymbol{u}$ *and* $\boldsymbol{i}$ *are rational, and the stationarity polynomial* (17) *is, up to a constant factor,*

$$H(z) = 7z^4 - 309z^3 + 2541z^2 - 8323z + 8820. \tag{24}$$

H is irreducible over the rationals and its Galois group is S_4. Exactly one of its roots is an interior stationary shape with $\Gamma^* > 0$, and the reduction Δ there exceeds the reductions at every boundary, so the best single branch exists, is unique, and is $z = 1.98670$, $L = 1.02209$, $C = 0.49247$ in normalized units. Consequently no formula that combines the inner products of the waveforms by sums, products and quotients, nor one that adds nested square roots, can give the optimal branch for every load, although the quartic itself is solvable by radicals.

Theorem 4 ($m = 4$)**.** *Let* $u(\tau) = \sqrt{2}\,(8\cos^4\tau + 4\cos^3\tau - 6\cos^2\tau - 2\cos\tau)$*, for which* $\mu_u(\xi) = \xi^4 - 30\xi^3 + 273\xi^2 - 820\xi + 576$*, and* $h(\xi) = (103\xi^3 - 2882\xi^2 + 22327\xi - 39708)/2880$*. The stationarity polynomial* (17) *is, up to a constant factor,*

$$\begin{aligned} f(z) = {} & 35z^7 + 1003z^6 - 60975z^5 + 1041441z^4 \\ & - 8601960z^3 + 36926304z^2 \\ & - 80912600z + 68173952, \end{aligned} \tag{25}$$

irreducible over the rationals with Galois group S_7, and the best single branch is an interior stationary shape, $z = 2.00242$, $L = 0.14215$, $C = 3.5133$ in normalized units. Consequently no formula by radicals in the inner products of the waveforms gives the optimal single branch for every load.

Both proofs are computations in exact rational arithmetic on the stated waveforms: irreducibility and the Galois groups follow from the factorization patterns of H and f modulo small primes, the uniqueness of the active root of H from a Sturm sequence, and the existence of an interior optimum from the comparison of Δ at an interior point with the limiting reductions at the roots of μ_u and at the one-element limits (Section S-IV of the Supplementary Material). A rational formula applied to rational data would give a rational root of H; a root obtainable through nested square roots has a normal closure of degree a power of two, whereas the normal closure of a root of the irreducible H is its splitting field of degree 24; and S_7 is not solvable. Fig. 1(b) shows the source current of the two loads as a function of the tuning: in both, the global optimum lies between the two lowest roots of μ_u, although positive improvement also occurs in other intervals. The optimum is at the irrational tuning given by the theorems, close to but above the bound (6). The two theorems state what a formula cannot do, and nothing more. They do not exclude an exact representation of the optimum as an algebraic number, nor a time-domain algorithm that isolates the root of (17) by a Sturm sequence and refines it, nor closed forms for particular classes of loads, such as loads with $m = 2$ satisfying the positivity criterion, or higher-dimensional loads whose $\boldsymbol{q}^*$ satisfies both compatibility and positivity. The degree of (17) proves nothing by itself: a reducible polynomial of degree four or seven may have a rational root. Table I collects the four cases.

TABLE I
THE SINGLE SERIES BRANCH ACCORDING TO THE REACTIVE DIMENSION

m	Branch drawing $\boldsymbol{q}^*$	Element values
1	C or L according to sign; series family	not unique
2	one branch iff $C_\parallel, \Gamma_\parallel, K$ nonzero, same sign	rational, (19)
3	generic target incompatible	$\deg F \le 4$; no universal rational form
4	generic target incompatible	$\deg F \le 7$; no universal radical form

IV. FREE-ORDER SYNTHESIS AND BANKS

Section III gives a complete criterion for a single branch to draw $\boldsymbol{q}^*$ when $m = 2$; for $m \ge 3$, this remains possible for targets satisfying compatibility and positivity. This section shows that a network of unrestricted order always does, constructs it from the two time-domain data of Section II, the polynomials μ_u and h of (7), and extends the geometry to banks of fixed branches and to switched banks.

A. A Positive Network for Every Compensable Current

By Lemma 1(c), every element of $\mathcal{Q}_u$ is $\mathsf{D}h(\mathsf{A})\boldsymbol{u}$ for a polynomial h of degree less than m. The construction has two steps: an admittance operator that draws the current and carries one free parameter, and a matrix test on that parameter that makes the admittance a reactance function.

Theorem 5 (Synthesis)**.** *Let $\boldsymbol{q} = \mathsf{D}h(\mathsf{A})\boldsymbol{u}$ with $\deg h < m$. Define $r(\xi) = \xi\mu_u'(\xi)/m$, $n = \operatorname{rem}(r\,h, \mu_u)$ and, for a scalar α,*

$$F_\alpha(\xi) = \frac{n(\xi) + \alpha\,\mu_u(\xi)}{r(\xi)}, \qquad \mathsf{Y}_\alpha = \mathsf{D}\,F_\alpha(\mathsf{A}). \tag{26}$$

(a) $\mathsf{Y}_\alpha\boldsymbol{u} = \boldsymbol{q}$ *for every α.*

(b) *Let $H_0 = -\operatorname{Bez}(r, \mu_u)$ and $B_n = \operatorname{Bez}(r, n)$ be the Bezoutian matrices of order m, defined by $[r(x)b(y) - r(y)b(x)]/(x-y) = \boldsymbol{v}(x)^{\mathsf{T}} \operatorname{Bez}(r,b)\,\boldsymbol{v}(y)$ with $\boldsymbol{v}(x) = (1, x, \dots, x^{m-1})^{\mathsf{T}}$. Then H_0 is positive definite, and whenever $\alpha \ge 0$ and $B_n - \alpha H_0 \preceq 0$ the admittance $Y_\alpha(s) = s\,F_\alpha(-s^2)$ is a reactance function with the Foster form*

$$Y_\alpha(s) = \alpha s + \frac{\gamma_0}{s} + \sum_{j=1}^{m-1} \frac{\gamma_j\, s}{s^2 + \mu_j}, \qquad \gamma_j \ge 0, \tag{27}$$

realized by a capacitor α, an inductor $1/\gamma_0$ and $m-1$ series branches with $L_j = 1/\gamma_j$, $C_j = \gamma_j/\mu_j$, at most $2m$ elements, omitting every zero-coefficient term.

(c) *Such an α exists: with $H_0 = RR^{\mathsf{T}}$ and $M = R^{-1}B_nR^{-\mathsf{T}}$,*

$$\alpha_G = \max\Big(0,\ \max_j \Big[M_{jj} + \sum_{k\ne j} |M_{jk}|\Big]\Big) \tag{28}$$

satisfies the condition of (b).

Part (a) holds because $rh - n$ is a multiple of μ_u, $\mu_u(\mathsf{A})\boldsymbol{u} = \boldsymbol{0}$ and $r(\mathsf{A})$ is invertible on $\mathcal{E}_u$, the roots of r being 0 and the roots of μ_u', which interlace those of μ_u. For (b), in the coordinates of evaluation at the roots μ_j of r both Bezoutian forms are diagonal, so $B_n - \alpha H_0 \preceq 0$ is equivalent to the residues $\rho_j = [n(\mu_j) + \alpha\mu_u(\mu_j)]/r'(\mu_j)$ being nonpositive, and $H_0 \succ 0$ to $\mu_u(\mu_j)/r'(\mu_j) < 0$, which holds by the interlacing; the partial fractions of F_α then give (27) with $\gamma_j = -\rho_j$, and each term is positive real. Part (c) is Gershgorin's bound (Section S-V of the Supplementary Material).

Three remarks. First, the construction uses only the coefficients of μ_u and h: no root of μ_u and no frequency is computed, and the elements can be extracted as a Cauer ladder by polynomial divisions without computing the poles μ_j; when the Foster form is used, the poles are the roots of r, that is, 0 and the roots of μ_u', a choice fixed by the voltage. Second, the operator $\alpha\,\mathsf{D}\mu_u(\mathsf{A})/r(\mathsf{A})$ draws no net terminal current at the design voltage, so every $\alpha \ge \alpha_G$ gives a different positive network with the same source current; moving the poles gives the larger family counted in [16], with $2m$ parameters and m conditions. The free parameters change the stored energy and the element stresses, not the source current. Third, together with Proposition 1 the theorem shows that $\mathcal{Q}_u$ is exactly the set of currents that lossless networks can draw, which completes the answer to the first question of the Introduction.

B. The Freedom of the Network for $m = 2$

For $m = 2$ the family of networks that draw $\boldsymbol{q}^*$ with one shunt capacitor and one series branch can be written in closed form, which makes the freedom visible.

Proposition 4. *Let $m = 2$ with the closure* (4) *and the coordinates $C_\parallel$, $\Gamma_\parallel$ of $\boldsymbol{q}^*$. Assume $\Gamma_\parallel \ne 0$. For every μ_1 between the two roots of μ_u, so that $\mu_u(\mu_1) < 0$, the network formed by a capacitor C_p in parallel with a series branch (L, C),*

$$C_p = C_\parallel - \frac{\Gamma_\parallel\,\mu_1}{p}, \qquad L = \frac{p}{\Gamma_\parallel\,\mu_u(\mu_1)}, \qquad C = \frac{\Gamma_\parallel\,\mu_u(\mu_1)}{p\,\mu_1}, \tag{29}$$

draws $\boldsymbol{q}^*$. The branch is positive if and only if $\Gamma_\parallel < 0$, and then $C_p \ge 0$ for every μ_1 when $C_\parallel \ge 0$ and for $\mu_1 \ge pC_\parallel/\Gamma_\parallel$ when $C_\parallel < 0$; at $C_p = 0$ the network is the single branch of Theorem 2. The dual family, a shunt inductor with a series branch, covers $C_\parallel < 0 < \Gamma_\parallel$.

The proof writes the branch current with Proposition 3, for which $\boldsymbol{W}(z) = z\,\mathsf{D}\boldsymbol{u} + p\,\mathsf{J}\boldsymbol{u}$ when $m = 2$, and equates coordinates in the frame $\{\mathsf{D}\boldsymbol{u}, \mathsf{J}\boldsymbol{u}\}$ (Section S-VI of the Supplementary Material).

The load of [17] has $C_\parallel > 0 > \Gamma_\parallel$, so (29) is positive on the whole interval between the two roots of μ_u, and the three-element network published there agrees, within its printed rounding, with the member at $\mu_1 = 1/(LC)$, obtained in [17] by solving the differential equation of the branch for the prescribed current. Fig. 2 shows the mean stored energy along the family: all members leave the same source current, and the energy varies by more than a factor of sixty, with a minimum in the interior and unbounded growth at the two ends, where the branch approaches resonance with a component of the voltage. The design of [27], which fixes the pole between the

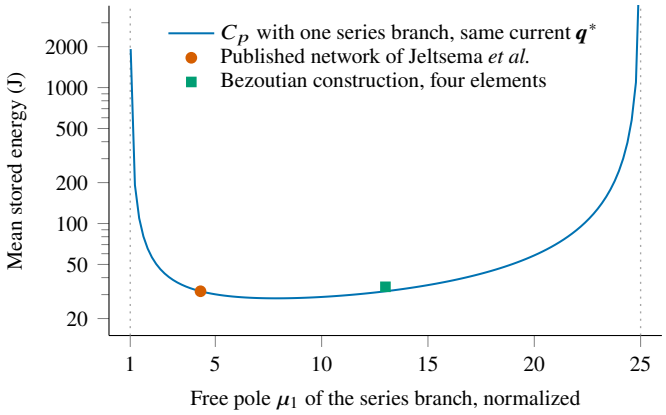


Fig. 2. Mean stored energy of the networks (29) for the R–L load of [17], as a function of the free pole μ_1 in normalized time. Every member draws the same current $\boldsymbol{q}^*$ and leaves the same source current; the energy varies by more than a factor of sixty across the family and grows without bound at the two roots of μ_u (dotted lines). The published three-element values approximate the member $\mu_1 = 1/(LC)$. The square is the four-element network of Theorem 5 with α chosen above α_G by the numerical positivity margin, whose pole is the root of μ_u'.

two frequencies and solves a linear program for the element values, is also a member, with a stored energy near the minimum; the four-element network of Theorem 5, whose pole is the root of μ_u' and whose fourth element is the inductor $1/\gamma_0$, stores slightly more. The optimal current is unique; the network, its element count and its stored energy are not, and the free pole is the parameter by which a designer trades them.

C. Banks of Fixed Branches

When the branches have fixed tunings, their currents have fixed shapes and the bank draws a current in a convex cone of $\mathcal{Q}_u$. Let $\boldsymbol{v}_1, \dots, \boldsymbol{v}_s \in \mathcal{Q}_u$ be the currents of the branches at unit scale, for instance $\boldsymbol{w}_{z_j}$ for a dictionary of tunings z_j or the pair $\mathsf{D}\boldsymbol{u}$, $\mathsf{J}\boldsymbol{u}$, and let $\mathcal{C} = \{\sum_j a_j \boldsymbol{v}_j : a_j \geq 0\}$.

Proposition 5. *The minimizer $\boldsymbol{q}_\mathcal{C}$ of $\|\boldsymbol{i} + \boldsymbol{q}\|$ over $\mathcal{C}$ exists and is unique. If S is the set of branches active in it, with independent $\boldsymbol{v}_j$ and blade $\boldsymbol{B}_S = \bigwedge_{j\in S} \boldsymbol{v}_j$, then*

$$\boldsymbol{q}_\mathcal{C} = -(\boldsymbol{i} \rfloor \boldsymbol{B}_S)\boldsymbol{B}_S^{-1}, \qquad a_j = -\boldsymbol{i}\cdot\boldsymbol{v}^j \geq 0 \quad (j \in S), \tag{30}$$

with $\{\boldsymbol{v}^j\}$ the reciprocal frame of $\{\boldsymbol{v}_j\}_{j\in S}$, and the residual $\boldsymbol{r} = \boldsymbol{i} + \boldsymbol{q}_\mathcal{C}$ satisfies

$$\boldsymbol{r}\cdot\boldsymbol{v}_j = 0 \quad (j \in S), \qquad \boldsymbol{r}\cdot\boldsymbol{v}_j \geq 0 \quad (j \notin S). \tag{31}$$

Conversely, a face S and amplitudes satisfying (30) *and* (31) *give the minimizer.*

These are the Karush–Kuhn–Tucker conditions of the projection onto a closed convex cone, written with the blade and the reciprocal frame of the active face; the projection onto the cone is piecewise linear, one projector per face, and is not obtained by clipping the unconstrained coordinates. When the active generators are dependent the current is still unique but the amplitudes are not, and no blade is inverted. The fixed-pole designs of [3], [27] are projections of this kind; when the dictionary has a positive null vector, the linear program of [27] finds non-negative amplitudes that draw the full projection, as shown in Section S-VI of the Supplementary Material.

Two operations on an existing bank follow from the same formulas. Adding a branch while keeping the others replaces $\boldsymbol{i}$ by the current residual $\boldsymbol{r}_k$ in (14): the new branch has $\Gamma(z) = [-\boldsymbol{r}_k\cdot\boldsymbol{w}_z]_+/\boldsymbol{w}_z^2$, improves the squared source current by exactly $\Delta_k(z) = [-\boldsymbol{r}_k\cdot\boldsymbol{w}_z]_+^2/\boldsymbol{w}_z^2$, which is positive if and only if $\boldsymbol{r}_k\cdot\boldsymbol{w}_z < 0$, and its best tuning is given by Proposition 3 applied to $\boldsymbol{r}_k$. Readjusting the amplitudes of the existing branches when a branch of free tuning is added uses the same polynomials with the projector onto the active face removed, $\boldsymbol{r} = (1 - \mathrm{proj}_{\boldsymbol{B}_S})\boldsymbol{i}$ and $\boldsymbol{W}_\perp(z) = (1 - \mathrm{proj}_{\boldsymbol{B}_S})\boldsymbol{W}(z)$, followed by the positivity and (31) checks (Section S-VI of the Supplementary Material). The sequence of additions improves monotonically; it does not find the best bank with all tunings free, which is a non-convex problem. For $m = 2$ the sequence closes in two steps: if the first bank is the positive parallel pair of Section III-A and compensable current remains, the projection of the residual has $C_\parallel$, $\Gamma_\parallel$ and K negative and Theorem 2 gives one positive series branch that completes it.

D. Switched Banks: States and the Instant of Connection

When fixed branches are connected or disconnected by thyristors, the admissible currents form a finite set, one periodic current $\boldsymbol{q}_\sigma$ for each state σ of the switches. Each $\boldsymbol{q}_\sigma$ is a sum of the branch currents of Proposition 5 with fixed amplitudes, so the cost $\|\boldsymbol{i} + \boldsymbol{q}_\sigma\|^2$ of every state is a quadratic form in the inner products of $\boldsymbol{i}$ with the branch waveforms and of the branch waveforms among themselves, computed once per period; the supervisor selects the admissible state of least cost, with hysteresis and a minimum dwell time (Section VI). The selection of a state by comparing costs is the procedure of [2], [7]; what the geometry adds is the cost of each state from the projection formulas, without iteration, and a criterion for the instant of connection.

That criterion uses the state of a branch as an element of the geometric algebra of the plane, $\mathrm{Cl}_{2,0}$, with the idempotents $P_\pm = (1 \pm e_1)/2$ and the nilpotents $N_\pm = (e_2 \pm e_{12})/2$, which satisfy $P_\pm^2 = P_\pm$, $N_\pm^2 = 0$, $N_+N_- = P_+$ and $N_-N_+ = P_-$. The state of a branch with current i_j and capacitor voltage v_j is

$$\begin{aligned} \boldsymbol{\psi}_j &= \sqrt{L_j}\, i_j\, P_+ + \sqrt{C_j}\, v_j\, N_-, \\ \dot{\boldsymbol{\psi}}_j &= -\Omega_j e_{12}\boldsymbol{\psi}_j + \frac{u(t)}{\sqrt{L_j}} P_+, \end{aligned} \tag{32}$$

with instantaneous signals $u(t)$, $i_j(t)$ and $v_j(t)$, and $\Omega_j = 1/\sqrt{L_jC_j}$; the two couplings of the branch are nilpotent and their combination is the rotation generated by e_{12}, and the stored energy is the scalar part $\langle\widetilde{\boldsymbol{\psi}_j}\boldsymbol{\psi}_j\rangle_0 = \frac{1}{2}L_j i_j^2 + \frac{1}{2}C_j v_j^2$. Products by the left in this representation compose and invert the operators of the branch, which is what a bivector acting by contraction does not do. For a candidate instant of connection, let $\boldsymbol{\psi}_j^*$ be the periodic orbit of the target state and $\delta\boldsymbol{\psi}_j = \boldsymbol{\psi}_j - \boldsymbol{\psi}_j^*$ the mismatch of the predicted state with respect to it; the mismatch energy

$$\mathcal{E}_\delta = \sum_j \langle\widetilde{\delta\boldsymbol{\psi}}_j\, \delta\boldsymbol{\psi}_j\rangle_0, \tag{33}$$

extended with the terms of the source inductance and of the fixed capacitor when present, is non-increasing for a fixed topology, linear passive components and repeated excitation: its derivative is minus the mismatch dissipation, and it is constant in the ideal lossless case, and bounds the excursions of every connected branch,

$$|v_j(t)| \le V_j^* + \sqrt{2\mathcal{E}_\delta / C_j}, \qquad |i_j(t)| \le I_j^* + \sqrt{2\mathcal{E}_\delta / L_j}, \quad (34)$$

where V_j^* and I_j^* are the peaks of the periodic orbit. The supervisor evaluates $\mathcal{E}_\delta$ at the instants of the next period, admits those for which (34) and a limit on di/dt hold, fires at the admitted instant of least $\mathcal{E}_\delta$, and repeats the check with the measured state at the moment of firing. Unlike the classical rule for thyristor-switched capacitors, which fires when the capacitor voltage equals the line voltage, the criterion weighs the capacitor voltage and the inductor current of the branch against its periodic orbit; Section VI compares both on the same manoeuvres. The bounds (34) are conservative and conditional on the excitation being repeated; they are admission rules for a connection, not a protection against arbitrary later disturbances.

V. Quantitative Comparison With Published Designs and Methods

The designs of Sections III and IV are compared with the published alternatives on eight cases: the example of Section III-C ($m = 2$); the load of Theorem 3 with $G = 0.8$ ($m = 3$); a 230 V, 1000 W load under a voltage with 3 % of second and 3 % of third harmonic ($m = 3$); the R–L load of [17] ($m = 2$); the load of [15] ($m = 3$), taken once with the exact elements of its Fig. 1 and once with the admittances printed in its eq. (12), which differ in the fourth decimal; and the load of [16] ($m = 3$), with an ideal source and with the source impedance of its example. Thirteen methods are applied to each case: no compensator; the capacitor (11); the parallel pair of Section III-A on its cone; the series branch of Theorem 2, when $m = 2$; a series branch by direct multistart search over $(\log L, \log C)$; the same search with the mean stored energy bounded by the smallest energy among the free-order networks of the case; a series branch with L fixed to a set of values and C optimized, following the fixed-inductance design principle of [16], without reproducing its specific iteration; Theorem 5 in Cauer and Foster forms; a Foster network with pole frequencies offset by 20 % from the excited frequencies, with the side chosen from the required susceptance; a network with poles 5 % away from the frequencies of the voltage, the rule of [3]; fixed intermediate poles with a linear program for positivity, the principle of [27]; and, when the source has impedance, a capacitor optimized in the coupled circuit. Currents are computed in the time domain as forced periodic responses: at fixed terminal voltage for ideal sources, and from the coupled circuit when source impedance is included; peak voltages and currents are periodic peaks; an independent audit reproduces the RMS values within a relative error of 10^{-6}. The nine published networks are evaluated with their printed component values. The full tables are in Section S-VII of the Supplementary Material; Table II and Fig. 3 show three cases.

Published results are reproduced. The three-element network of [17] leaves 2.0006 A with a current efficiency index of 0.905, the value reported there; the network of Fig. 3 of [15] leaves 3.097 A against the 3.10 A printed; the Foster network of [16], evaluated with its source impedance, leaves 49.73 A against 49.7 A, and the entries of its Table I are reproduced within 0.2 A except the column $L = 0.20$ H, whose printed total current is inconsistent with its own fundamental and is recomputed as 49.9 A.

Same projection, different realizations. In the fixed-voltage cases the five free-order constructions leave the same source current, as Theorem 1 requires, with three to six elements and mean stored energies that differ by an order of magnitude: from 28 J with the fixed-pole linear program to 414 J with poles near the frequencies for the load of [17]; from 4.8 J to 38 J, with peak element voltages from 364 V to 3498 V, for the 230 V case; from 2813 J to 8063 J for the load of [15], whose published four-element network stores 1817 J. The rule that places poles next to the frequencies produces the largest energies and voltages; this is a consequence of the rule, not of the projection, and it is what the free parameters of Section IV-B let a designer avoid.

One branch against free order. In the $m = 3$ cases compared here, the best single branch does not attain $\boldsymbol{q}^*$; this is not a universal impossibility for $m = 3$. For the load of [15] the best branch, a certified global minimum with $L = 10.23$ H and $C = 0.0433$ F, leaves 7.502 A, more than twice the bound of 3.097 A reached with five or six elements; the procedure with fixed L leaves 7.538 A. For the 230 V case the energy-bounded branch leaves 4.350 A against the bound of 4.348 A: there, one branch is almost enough. For Jeltsema's $m = 2$ load, Theorem 2(ii) predicts an unattained boundary infimum. An independent exact audit establishes the same classification for the $m = 3$ ideal-source case of Czarnecki 1987: the unconstrained search approaches the resonant infimum, 2.061 A for [17] and 50.01 A for the ideal-source case of [16], the finite numerical approximations store $4 \cdot 10^5$ J and $1.4 \cdot 10^7$ J and have peaks above 10^5 V, so they are not practical designs; with the energy bounded, the best branch leaves 2.585 A and 51.11 A, and the fixed-L procedure 2.071 A and 50.35 A. For the example with $m = 2$ and concordant signs, the formula (19), the direct search and all free-order networks agree on 0.4932 A; Theorem 5 draws the same current with four elements and 0.49 mJ against 0.90 mJ for the single branch.

Source impedance. With the source of the example of [16], the capacitor (11) computed with the initial terminal voltage resonates with the source inductance and raises the source current from 95.5 A to 227.5 A, which is the warning of [16] in the present terms and the reason why (H1) is a hypothesis and not a detail; a capacitor optimized in the coupled circuit leaves 66.5 A, and the free-order networks of the fixed-voltage design, evaluated in the coupled circuit, leave 49.74 A. The projection at fixed voltage is not an optimality statement for the coupled circuit, and the near-equal currents of the tested free-order networks do not certify coupled-circuit optimality.

TABLE II
SOURCE CURRENT i_s (A), NUMBER OF ELEMENTS n AND MEAN STORED ENERGY E (J) OF THE DESIGNS ON THREE CASES, FORCED PERIODIC RESPONSE AT FIXED TERMINAL VOLTAGE

	(a) R–L load [17]			(b) Load of [15]			(c) 230 V, 4.24 % THD		
Design	i_s	n	E	i_s	n	E	i_s	n	E
No compensator	4.500	0	–	12.241	0	–	6.958	0	–
Capacitor, (11)	4.058	1	4.52	12.241	0	–	4.408	1	1.96
One branch, energy bounded	2.585	2	28.4	7.502	2	1269	4.350	2	4.83
One branch, global minimum	–	–	–	7.502	2	1269	–	–	–
One branch, L fixed [16]	2.071	2	418	7.538	2	1337	4.348	2	8.19
Theorem 5, Cauer	2.001	4	34.2	3.097	6	2813	4.348	6	9.64
Fixed poles, linear program [27]	2.001	3	28.4	3.097	5	3301	4.348	5	4.83
Foster, prescribed 20 % offsets	2.001	4	111	3.097	6	3224	4.348	6	18.1
Poles near the frequencies [3]	2.001	4	414	3.097	6	8063	4.348	6	37.8
Published network	2.001	3	31.8	3.097	4	1817	–	–	–

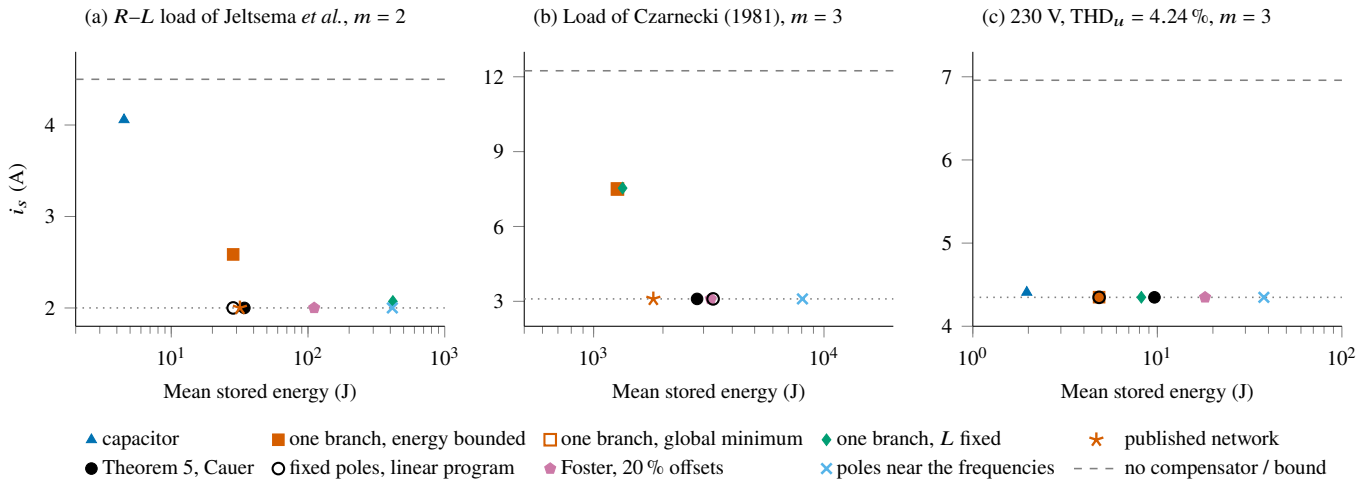


Fig. 3. Source current against mean stored energy for the designs of Table II. Dashed line: no compensator; dotted line: the bound (6), attained by every free-order network. (a) R–L load of [17]: the unattained infimum of the single branch (Theorem 2(ii)) is not plotted; the energy-bounded branch and the branch with fixed L stay above the bound. (b) Load of [15]: the single branch has a finite global minimum, far from the bound; the published four-element network stores less energy than the automatic realizations. (c) 230 V with second and third harmonics each at 3 % (voltage THD 4.24 %): one branch is almost sufficient; the realizations differ by a factor of eight in energy.

VI. SIMULATION STUDY OF A SWITCHED BANK

A. Model and Supervisor

The closed-loop study uses a 50 Hz source of 230 V RMS, either with equal first, second and third harmonic amplitudes (the test waveform) or with second and third harmonics each at 3 % of the fundamental (the moderate waveform). The load has a 1000 W nominal reference and is an exogenous current waveform in the dynamic runs; it has no internal states. The bank is based on the three-element realization of the project $m = 3$ load at normalized tuning $z = 27/13$, not the $m = 2$ family of Proposition 4. At the 230 V scale it has $C_p = 2.8929$ μF and a full-admittance series branch with $L = 130.09$ mH and $C = 37.500$ μF; the second branch has twice this inductance and half this capacitance. A fixed capacitor and the two switchable branches contain five reactive elements and give four states. Series resistance and ESR correspond to $Q_L = 100$ and $Q_C = 500$ at 50 Hz; the bleeders are 42.32 kΩ and 84.64 kΩ for the full and half branches. The idealized valves close at the admitted instant and open at natural current zero. The source is ideal or has $(R_g, L_g) = (0.4\ \Omega, 0.8\ \text{mH})$ or $(1.5\ \Omega, 3\ \text{mH})$; with impedance, the terminal voltage is determined by the coupled circuit. Closed-loop measurements are noiseless. The network-aware variant uses the known source impedance; the earlier terminal-cost policy is reported separately.

Each period the supervisor forms the inner products of the measured voltage and load current with the branch waveforms, evaluates the cost of the four states by Section IV-D, and requests at most one branch change per decision, with hysteresis of 0.025 A^2 and a dwell of 60 ms; the admission stage uses the measured branch states and a 2500 A/s slope limit with a 1 % planning margin; a connection is executed at the instant of the next period that minimizes the mismatch energy (33) among those that satisfy the limits (34), and is re-checked with the measured state when the instant arrives. This supervisor is labelled Projection. It is compared with a Fundamental supervisor that uses only the fundamental reactive current, with the four fixed states, and with the best fixed state chosen after the run, which is a comparator and not a causal policy.

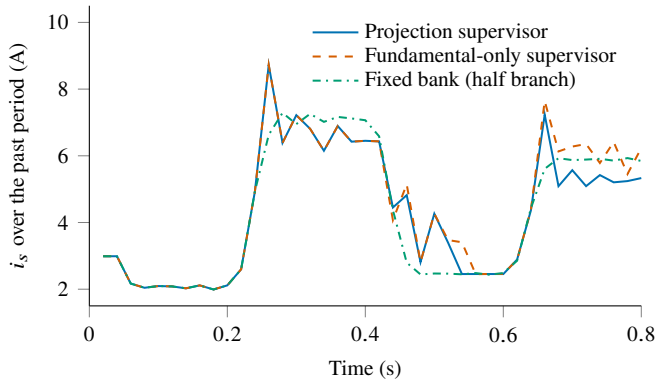


Fig. 4. Closed-loop run of 0.8 s with the test waveform at 230 V and a load that changes during the run: RMS source current over each past period for the Projection and the Fundamental supervisors and for the fixed state with one branch. Over the run, the Projection supervisor leaves 4.712 A and the Fundamental one 4.914 A, with nine manoeuvres each and no rejected connection; the fixed state leaves 4.737 A.

B. Results

Fig. 4 shows the closed-loop run with the test waveform and a changing load: the Projection supervisor follows the prescribed load profile subject to its dwell and admission rules, and its current over the run is 4.1 % below that of the Fundamental supervisor, which ignores the components of the reactive current outside the fundamental. Table III collects the steady-state and closed-loop results with the moderate waveform. With this waveform the two supervisors take the same decisions, and the projection supervisor stays within 3.7 %, 1.2 % and −2.4 % of the best fixed state for the ideal source and the two impedant sources. Including the source impedance in the predictor does not change the decisions in these runs, but the admission limits do: with 1100 V and 8 A the supervisor executes one manoeuvre instead of three, the peak branch current halves, and the source current is lower; with a 900 V limit no connection is admitted. The tuning results illustrate the current, stress and loss trade-off for the $m = 3$ bank: among the sampled family members, the three-element reference gives the lowest losses on this waveform but starts with a capacitor peak of 1020 V, while a dictionary of branches detuned by 14 % and fitted to the moderate waveform leaves almost the same current with one fifth of the losses and half the startup peak; the same dictionary under the test waveform leaves 11.71 A, above the 6.30 A of the uncompensated load, so the tuning must follow the actual waveform. In the separate unscaled connection experiment ($\sqrt{3}$ V RMS), the mismatch-energy criterion gives the lowest peaks of current and capacitor voltage, at the price of a longer wait and a larger initial di/dt than the classical rule of equal valve voltage; neither criterion dominates in every metric.

C. Estimation From Noisy Records and Computational Cost

The non-negative bank coefficients at a fixed tuning were estimated from eight synthetic noisy records (two seeds at each noise level, 0, 0.1, 1 and 3 %), using up to eight past periods. This experiment does not identify the closure polynomial in (H3). At 3 % noise, the coefficient error is 22.2–22.6 % from one period, 5.88–6.44 % after averaging eight periods, and 0.159–0.443 % with regularization. Parallel mSOGIs based on [35], with known 50/100/150 Hz support and no frequency-locked loop, give 0.229–0.660 % for $a = 40$ s^{-1} and 0.278–1.508 % for $a = 100$ s^{-1}. The largest excess of squared source current is $1.501 \cdot 10^{-2}$ A^2. These eight records illustrate the effect of preprocessing on a fixed-dictionary fit; they do not establish a general noise-robustness guarantee or identify unknown spectral support. With precomputed branch waveforms and state energies, each cost evaluation requires approximately $2NM + O(N)$ operations for correlations and $2M$ per state, hence $O(M2^M)$ for all binary states. MATLAB host timings are 1.2 μs for $(N, M) = (63, 2)$ and 4.8 μs for $(1023, 8)$, versus 6.2 μs and 57 μs using a full-window FFT. These timings exclude filtering, period detection, bank design, waveform updates and switch execution; they do not establish embedded-controller execution time (Section S-VIII of the Supplementary Material).

TABLE III
SIMULATION RESULTS: 230 V, 1000 W NOMINAL REFERENCE, EXCEPT FOR THE UNSCALED CONNECTION BLOCK

Closed loop, 0.8 s: RMS source current (A)			
Source	Projection	Best fixed	Excess
ideal	5.238	5.049	+3.7 %
0.4 Ω, 0.8 mH	5.090	5.029	+1.2 %
1.5 Ω, 3 mH	4.830	4.949	−2.4 %
Admission limits, source 0.4 Ω, 0.8 mH			
Limits	i_s (A)	I_{pk} (A)	Man.
none	5.103	14.0	3
1100 V, 8 A	5.029	7.1	1
900 V, 16 A	6.276	–	0
Fixed banks with losses, steady state			
Design	i_s (A)	P_{loss} (W)	$V_{C,\mathrm{start}}$ (V)
three-element $m = 3$ reference	4.417	15.9	1020
half-admittance branch	5.102	8.0	1020
dictionary, 14 % detuned	4.397	3.1	540
Unscaled connection ($\sqrt{3}$ V RMS, $R = 0.05$ Ω, $R_b = 1$ kΩ)			
Criterion	I_{pk} (A)	$V_{C,\mathrm{pk}}$ (V)	wait (ms)
immediate	3.690	5.32	0
equal valve voltage	3.931	5.79	1.5
mismatch energy (33)	2.927	4.16	18.9

VII. DISCUSSION AND LIMITATIONS

The results rest on the four hypotheses of Section II-A, and the simulations show where each one matters. The fixed terminal voltage (H1) is the one with the largest effect: a capacitor designed for the initial voltage can resonate with the source inductance, as the 227.5 A of Section V show, and the projection at fixed voltage is not an optimality statement for the coupled circuit; the designs remain close to the best found in the coupled circuit in the cases studied, and the supervisor can include a known source impedance in its predictor, but a design for a source of significant impedance has to be made on the coupled circuit. The finite closure (H3) is a model hypothesis assessed for numerical consistency in the sampled examples; the noise study concerns bank-amplitude fitting at fixed tuning and known support, and does not validate

identification of the closure or its minimal degree. The forced periodic response (H4) excludes the connection transient, which the mismatch energy bounds but does not remove: the reference design starts with a capacitor peak of 1020 V, and detuned dictionaries lower it at the price of a current that depends on the actual waveform. In the dynamic switching study the load is an exogenous current waveform, whereas Section V evaluates the stated LTI load admittances. Closed-loop interaction with a rectifier or other load with internal states is not simulated; the valves are ideal, the measurements are noiseless in the closed-loop runs, and no hardware has been built. The time-domain computation needs no spectral stage, and its operation count is small, but the reported host timings cover cost evaluation rather than execution of the complete controller on an embedded target.

On the theoretical side, the single branch for $m \geq 3$ is a root of a polynomial whose degree grows with m, and Theorems 3 and 4 show that this is not an artefact of the method; the general single-branch problem cannot be solved by a universal rational formula, and numerical optimization, root isolation or restricted dictionaries remain available, and the sequence of additions of Section IV-C is non-worsening but carries no guarantee of finding the best bank with all tunings free. The comparison does not include the genetic algorithm of [28], whose original code was not available. Multiphase circuits, voltages with a direct component and loads that change within a period are outside the scope of the present formulation.

VIII. Conclusion

Lossless shunt compensation of a periodic non-sinusoidal load has been formulated as the projection of the load current onto the subspace $\mathcal{Q}_u$ generated by the derivatives of the voltage waveform, in geometric algebra and in the time domain. The subspace depends on the voltage alone and is exactly the set of currents that lossless networks can draw; the optimal compensating current is the negative projection of the load current onto it; the capacitor, the reactor and the parallel pair of the classical time-domain designs are projections onto its subspaces; and for circuit laws linear in the unknown parameters, nondegenerate blade quotients recover a unique compatible parameter vector, whose positivity must be checked separately. For a voltage with two reactive directions, the series branch with both elements free has a rational closed form with a sign criterion for existence and uniqueness, supplemented by boundary comparisons when no finite branch is optimal, including the boundary behavior of the two-direction inductive example; examples with three and four directions exclude universal rational and radical formulas, respectively, while algebraic root representations and special-case closed forms remain possible. A network of positive elements that draws the optimal current always exists and is constructed from two polynomials computed from the waveform, with a Bezoutian criterion of positive realness and realization freedom that can substantially change stored energy without changing the design current; the same geometry gives the cone of a bank with fixed tunings, the finite set of states of a switched bank, and an energy criterion for the instant of connection. The comparison reproduces the published designs and separates attainable optima from resonant infima, and the closed-loop simulations show the supervisor working with losses, source impedance and admission limits. The next steps are the coupled-circuit design for sources of significant impedance, loads with internal states, and the measurement of the computation on a controller.

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Supplementary Material for "Optimal Passive LC Compensation Under Nonsinusoidal Voltage: Extended Version"

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***Abstract*—This document contains the proofs omitted from the paper, the matrix forms of its geometric expressions, and the exact data of the two algebraic examples. The notation is that of the paper: waveforms are vectors x of the space $\mathcal{H}$ of periodic waveforms with the mean-square inner product $\langle x, y\rangle$, $x = \|x\|$ is the RMS value, D is the derivative, $\mathsf{J} = \mathsf{D}^{-1}$ the zero-mean periodic antiderivative, $\mathsf{A} = -\mathsf{D}^2$, $\mathcal{E}_u$ the cyclic subspace of the voltage, $\mathcal{Q}_u = \mathsf{D}\mathcal{E}_u$ the compensable subspace, μ_u the minimal polynomial of A on u and m its degree.**

S-I. Proofs of Section II of the Paper

A. Lemma 1

Under the finite-closure hypothesis (H3): (a) μ_u has m distinct positive roots; (b) $\mathcal{E}_u \perp \mathcal{Q}_u$; (c) $\mathsf{J}\boldsymbol{u} \in \mathcal{Q}_u$, and $\mathcal{Q}_u = \operatorname{span}\{\mathsf{D}\boldsymbol{u}, \mathsf{D}^3\boldsymbol{u}, \ldots, \mathsf{D}^{2m-1}\boldsymbol{u}\}$ has dimension m.

Proof: (a) $\mathcal{E}_u$ is finite dimensional and invariant under A, which is self-adjoint and positive definite on it, because $\langle \mathsf{A}\boldsymbol{x}, \boldsymbol{x}\rangle = \|\mathsf{D}\boldsymbol{x}\|^2 > 0$ for every non-zero zero-mean $\boldsymbol{x}$. Since $\boldsymbol{u}$ is a cyclic vector of the restriction, the minimal polynomial μ_u of $\boldsymbol{u}$ coincides with the characteristic polynomial of the restriction, whose eigenvalues are positive; a self-adjoint operator with a cyclic vector has simple spectrum, so the roots are distinct. (b) Because A is self-adjoint and commutes with D, $\langle \mathsf{A}^j\boldsymbol{u}, \mathsf{D}\mathsf{A}^k\boldsymbol{u}\rangle = \langle \boldsymbol{u}, \mathsf{D}\mathsf{A}^n\boldsymbol{u}\rangle$ with $n = j+k$, and $\langle \boldsymbol{u}, \mathsf{D}\mathsf{A}^n\boldsymbol{u}\rangle = (-1)^n\langle \boldsymbol{u}, \mathsf{D}^{2n+1}\boldsymbol{u}\rangle$. Moving n derivatives across the inner product, each with the sign of $\mathsf{D}^* = -\mathsf{D}$, gives $\langle \boldsymbol{u}, \mathsf{D}^{2n+1}\boldsymbol{u}\rangle = (-1)^n\langle \mathsf{D}^n\boldsymbol{u}, \mathsf{D}^{n+1}\boldsymbol{u}\rangle$, and $\langle \boldsymbol{y}, \mathsf{D}\boldsymbol{y}\rangle = 0$ for every $\boldsymbol{y}$ by skew-adjointness. (c) The constant term of μ_u is, up to sign, the product of its roots, non-zero by (a); hence $\boldsymbol{u}$ is a linear combination of $\mathsf{A}\boldsymbol{u}, \ldots, \mathsf{A}^m\boldsymbol{u}$. Applying J and using $\mathsf{J}\mathsf{A}^k\boldsymbol{u} = (-1)^k\mathsf{D}^{2k-1}\boldsymbol{u}$ for $k \geq 1$ gives $\mathsf{J}\boldsymbol{u} \in \mathcal{Q}_u$. The spanning set follows from $\mathsf{D}\mathsf{A}^k\boldsymbol{u} = (-1)^k\mathsf{D}^{2k+1}\boldsymbol{u}$, and $\dim \mathcal{Q}_u = \dim \mathcal{E}_u = m$ because D is injective on zero-mean waveforms. ∎

B. Proposition 2 (quotients of blades)

Let $\boldsymbol{u} = \sum_{k=1}^{n} \theta_k T_k \boldsymbol{q}$, with known linear operators T_k, and let $\boldsymbol{x}_k = T_k\boldsymbol{q}$ and $\boldsymbol{X} = \boldsymbol{x}_1 \wedge \cdots \wedge \boldsymbol{x}_n \neq \boldsymbol{0}$. A parameter vector satisfying the relation exists if and only if $\boldsymbol{u} \wedge \boldsymbol{X} = \boldsymbol{0}$, and it is then unique and given by $\theta_k = \boldsymbol{X}_k\boldsymbol{X}^{-1}$, where $\boldsymbol{X}_k$ is $\boldsymbol{X}$ with $\boldsymbol{x}_k$ replaced by $\boldsymbol{u}$.

Proof: If $\boldsymbol{u} = \sum_k \theta_k\boldsymbol{x}_k$, then $\boldsymbol{u} \wedge \boldsymbol{X} = \sum_k \theta_k \boldsymbol{x}_k \wedge \boldsymbol{X} = \boldsymbol{0}$ because each $\boldsymbol{x}_k$ appears twice in the outer product. By multilinearity, $\boldsymbol{X}_k = \sum_l \theta_l\, \boldsymbol{x}_1 \wedge \cdots \wedge \boldsymbol{x}_l \wedge \cdots \wedge \boldsymbol{x}_n$ with $\boldsymbol{x}_l$ in position k, and all terms with $l \neq k$ vanish by antisymmetry, so $\boldsymbol{X}_k = \theta_k\boldsymbol{X}$ and $\boldsymbol{X}_k\boldsymbol{X}^{-1} = \theta_k$. Conversely, $\boldsymbol{X} \neq \boldsymbol{0}$ means that the $\boldsymbol{x}_k$ are linearly independent, and $\boldsymbol{u} \wedge \boldsymbol{X} = \boldsymbol{0}$ means that $\boldsymbol{u}$ lies in their span, so the coefficients exist and are unique. ∎

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S-II. Proof of Proposition 3 and Matrix Counterparts

A. Proposition 3 (finite algebraic reduction)

Let $\mu_u(\xi) = \sum_{j=0}^{m} \pi_j \xi^j$, $\pi_m = 1$, and $\boldsymbol{W}(z) = \sum_{j=1}^{m} \pi_j \sum_{k=0}^{j-1} z^{j-1-k}\mathsf{A}^k\mathsf{D}\boldsymbol{u}$.

Proof: For every $j \geq 1$, $(z - \mathsf{A})\sum_{k=0}^{j-1} z^{j-1-k}\mathsf{A}^k = z^j - \mathsf{A}^j$, a telescoping sum. Hence $(z - \mathsf{A})\boldsymbol{W}(z) = \sum_{j=1}^{m} \pi_j(z^j - \mathsf{A}^j)\mathsf{D}\boldsymbol{u} = \mu_u(z)\mathsf{D}\boldsymbol{u} - \mu_u(\mathsf{A})\mathsf{D}\boldsymbol{u}$, and $\mu_u(\mathsf{A})\mathsf{D}\boldsymbol{u} = \mathsf{D}\mu_u(\mathsf{A})\boldsymbol{u} = \boldsymbol{0}$. When $\mu_u(z) \neq 0$, the restriction of $z - \mathsf{A}$ to $\mathcal{Q}_u$ is invertible and $\boldsymbol{w}_z = \boldsymbol{W}(z)/\mu_u(z)$. With $a(z) = \boldsymbol{i} \cdot \boldsymbol{W}(z)$ and $b(z) = \boldsymbol{W}(z)^2$, the best scale $\Gamma^*(z) = [-\boldsymbol{i} \cdot \boldsymbol{w}_z]_+/\boldsymbol{w}_z^2$ becomes $[-\mu_u(z)a(z)]_+/b(z)$ and the reduction $\Delta(z) = [-\boldsymbol{i} \cdot \boldsymbol{w}_z]_+^2/\boldsymbol{w}_z^2$ becomes $a(z)^2/b(z)$ when $\mu_u(z)a(z) < 0$ and zero otherwise; $b(z) > 0$ on the real axis because $\boldsymbol{W}(z) \neq \boldsymbol{0}$ there, μ_u being minimal. On the active set $a \neq 0$, $(a^2/b)' = aF/b^2$, so stationarity is equivalent to $F = 2a'b - ab' = 0$. If $a \equiv 0$, the best admitted choice is no branch and the root step is unnecessary. Since $\deg a \leq m - 1$ and $\deg b \leq 2m - 2$, the terms of degree $3m - 4$ in $2a'b$ and in ab' are both $2(m-1)a_{m-1}b_{2m-2}z^{3m-4}$ and cancel, so $\deg F \leq 3m - 5$. The vector identity $F(z) = -2\,\boldsymbol{i} \cdot [(\boldsymbol{W}(z) \wedge \boldsymbol{W}'(z))\boldsymbol{W}(z)]$ follows from $(\boldsymbol{a} \wedge \boldsymbol{b})\boldsymbol{a} = \boldsymbol{a}(\boldsymbol{a} \cdot \boldsymbol{b}) - \boldsymbol{b}\,a^2$ with $\boldsymbol{a} = \boldsymbol{W}$ and $\boldsymbol{b} = \boldsymbol{W}'$. The candidates for the global optimum are the positive roots of F with $\Gamma^* > 0$, the absence of branch, the limits $z \to 0$ and $z \to \infty$, in which $\boldsymbol{w}_z$ tends to multiples of $\mathsf{J}\boldsymbol{u}$ and of $\mathsf{D}\boldsymbol{u}$ and the branch becomes an inductor or a capacitor alone, and the one-sided limits at the roots of μ_u, where $\Gamma^* \to 0$ and the limiting current may remain nonzero on the active side. ∎

B. Matrix counterparts of the coordinates

With the moments and correlations

$$M_{-1}=\|\mathsf{J}\boldsymbol{u}\|^2,\qquad M_0=u^2,\qquad M_1=\|\mathsf{D}\boldsymbol{u}\|^2,$$
$$M_2=\|\mathsf{D}^2\boldsymbol{u}\|^2,\qquad \rho_{-1}=\langle\boldsymbol{i},\mathsf{J}\boldsymbol{u}\rangle,\qquad \rho_1=\langle\boldsymbol{i},\mathsf{D}\boldsymbol{u}\rangle, \tag{S1}$$

and $\delta=M_{-1}M_1-M_0^2=\|\mathsf{D}\boldsymbol{u}\wedge\mathsf{J}\boldsymbol{u}\|^2$, the coordinates of the parallel pair of the paper, Section III-A, are

$$C_\parallel=-\frac{M_{-1}\rho_1+M_0\rho_{-1}}{\delta},\qquad \Gamma_\parallel=-\frac{M_0\rho_1+M_1\rho_{-1}}{\delta}, \tag{S2}$$

which is the solution of the Gram system of the pair, $\langle\mathsf{D}\boldsymbol{u},\mathsf{J}\boldsymbol{u}\rangle=-M_0$ being its off-diagonal entry. For $m=2$ the coefficients of the closure $\mathsf{D}^4\boldsymbol{u}+s\mathsf{D}^2\boldsymbol{u}+p\boldsymbol{u}=\boldsymbol{0}$ are

$$s=\frac{M_{-1}M_2-M_0M_1}{\delta},\qquad p=\frac{M_0M_2-M_1^2}{\delta}, \tag{S3}$$

obtained from the inner products of the closure with $\mathsf{J}^2\boldsymbol{u}$ and with $\boldsymbol{u}$, that is, $M_1-sM_0+pM_{-1}=0$ and $M_2-sM_1+pM_0=0$. Equations (S2) and (S3) are the expansions of the reciprocal-frame expressions of the paper and involve the same six inner products.

S-III. Theorem 2: Degenerate Cases and Boundary

Let $m=2$, $\boldsymbol{q}^*=C_\parallel\mathsf{D}\boldsymbol{u}+\Gamma_\parallel\mathsf{J}\boldsymbol{u}$ and $K=pC_\parallel^2-sC_\parallel\Gamma_\parallel+\Gamma_\parallel^2$, with $\boldsymbol{x}=\mathsf{D}\boldsymbol{q}^*$, $\boldsymbol{y}=\mathsf{J}\boldsymbol{q}^*$ and $\boldsymbol{x}\wedge\boldsymbol{y}=-(K/p)\,\boldsymbol{u}\wedge\mathsf{D}^2\boldsymbol{u}$.

Degenerate cases. If $\boldsymbol{q}^*=\boldsymbol{0}$, the load current is orthogonal to $\mathcal{Q}_u$, every nonzero $\boldsymbol{q}\in\mathcal{Q}_u$ increases the source current, and no branch is used. If $\Gamma_\parallel=0$ and $C_\parallel>0$, $\boldsymbol{q}^*=C_\parallel\mathsf{D}\boldsymbol{u}$ is the current of the capacitor $C=C_\parallel$ alone, which is the limit $L\to0$ of the branch; if $C_\parallel=0$ and $\Gamma_\parallel>0$, $\boldsymbol{q}^*$ is the current of the inductor $L=1/\Gamma_\parallel$ alone, the limit $C\to\infty$. If $K=0$ and $\boldsymbol{q}^*\neq\boldsymbol{0}$, then $\boldsymbol{x}$ and $\boldsymbol{y}$ are parallel, so $\mathsf{A}\boldsymbol{q}^*=\lambda\boldsymbol{q}^*$ for a root λ of μ_u, and the branch law $\boldsymbol{u}=L\boldsymbol{x}+S\boldsymbol{y}$ would confine $\boldsymbol{u}$ to one eigen-direction of A, against $\boldsymbol{u}\wedge\mathsf{D}^2\boldsymbol{u}\neq\boldsymbol{0}$; no finite branch draws such a current.

Independence of $\boldsymbol{w}_z$ and $\partial_z\boldsymbol{w}_z$. In a basis of eigenvectors of A on $\mathcal{Q}_u$ with eigenvalues $\lambda_1\neq\lambda_2$, write $\mathsf{D}\boldsymbol{u}=d_1\boldsymbol{f}_1+d_2\boldsymbol{f}_2$ with $d_1d_2\neq0$, which is the condition $\boldsymbol{u}\wedge\mathsf{D}^2\boldsymbol{u}\neq\boldsymbol{0}$. Then $\boldsymbol{w}_z=\sum_k d_k(z-\lambda_k)^{-1}\boldsymbol{f}_k$ and $\partial_z\boldsymbol{w}_z=-\sum_k d_k(z-\lambda_k)^{-2}\boldsymbol{f}_k$, and the determinant of their coordinates is $d_1d_2(\lambda_1-\lambda_2)(z-\lambda_1)^{-2}(z-\lambda_2)^{-2}\neq0$ for every admissible z.

Boundary. Suppose the realizability criterion in Theorem 2(i) fails, including any zero coefficient, and let a positive finite branch (Γ,z) be a local minimizer of $\|\boldsymbol{i}+\Gamma\boldsymbol{w}_z\|$. Stationarity in Γ and in z makes the residual $\boldsymbol{i}+\Gamma\boldsymbol{w}_z$ orthogonal to $\boldsymbol{w}_z$ and to $\partial_z\boldsymbol{w}_z$, which span $\mathcal{Q}_u$; the residual is then orthogonal to $\mathcal{Q}_u$ and $\Gamma\boldsymbol{w}_z=-\operatorname{proj}_{\mathcal{Q}_u}\boldsymbol{i}=\boldsymbol{q}^*$, which no positive finite branch draws when the sign condition fails. Hence no interior local minimizer exists. The function $z\mapsto\|\boldsymbol{i}+\Gamma^*(z)\boldsymbol{w}_z\|$ is continuous on the open admissible set $\{z>0:\mu_u(z)\neq0\}$, so its infimum is approached at the boundary: as $z\to0$ or $z\to\infty$, where the branch becomes an inductor or a capacitor alone, or as z tends to a root of μ_u, where $\Gamma^*\to0$: on the active side the limiting current is the projection of $-\boldsymbol{i}$ onto the corresponding eigen-direction of A in $\mathcal{Q}_u$, and on the inactive side it is zero, which corresponds to $L\to\infty$ and $C\to0$ with $LC\to1/\lambda$; or $\Gamma^*=0$ on an interval, where no branch is used. No boundary limit is attained by a branch with both L,C strictly positive and finite. A pure-capacitor, pure-inductor or omitted-branch limit can be attained in the enlarged topology class.

S-IV. Theorems 3 and 4: Data and Proofs

Both examples are constructed in normalized time τ of period 2π with $\mathsf{D}=d/d\tau$, a load $\boldsymbol{i}=G\boldsymbol{u}+h(\mathsf{A})\mathsf{D}\boldsymbol{u}$ and $G>0$ arbitrary, which does not affect $\boldsymbol{q}^*$. All quantities are computed with exact rational arithmetic from the waveforms and their integrals over one period.

A. Theorem 3 ($m=3$)

The voltage $u(\tau)=\sqrt{2}\,(4\cos^3\tau+2\cos^2\tau-2\cos\tau-1)$ has zero mean and reactive dimension three, with $\mu_u(\xi)=\xi^3-14\xi^2+49\xi-36$. With $h(\xi)=(-13\xi^2+155\xi-322)/180$, the lossless part of the load is

$$h(\mathsf{A})\mathsf{D}\boldsymbol{u}=\tfrac{11}{20}\,\mathsf{D}\boldsymbol{u}+\tfrac{17}{10}\,\mathsf{J}\boldsymbol{u}+\tfrac{3}{4}\,(6-\mathsf{A})^{-1}\mathsf{D}\boldsymbol{u}, \tag{S4}$$

because the difference of the two sides, multiplied by $\mathsf{A}(6-\mathsf{A})$, equals $(13\mathsf{A}-51)\mu_u(\mathsf{A})\mathsf{D}\boldsymbol{u}/180=\boldsymbol{0}$, and $\mathsf{A}(6-\mathsf{A})$ is invertible on $\mathcal{Q}_u$ since $\mu_u(0)\neq0$ and $\mu_u(6)\neq0$; this is the current of a capacitor $11/20$, an inductor $10/17$ and a series branch with $L=4/3$, $C=1/8$, in parallel. In the basis $\{\mathsf{D}\boldsymbol{u},\mathsf{A}\mathsf{D}\boldsymbol{u},\mathsf{A}^2\mathsf{D}\boldsymbol{u}\}$ of $\mathcal{Q}_u$ the Gram matrix and the inner products with $\boldsymbol{i}$ are

$$\begin{pmatrix}14&98&794\\98&794&6818\\794&6818&60074\end{pmatrix},\qquad (2,\ 16,\ 112), \tag{S5}$$

so that $a(z)=2z^2-12z-14$, $b(z)=14z^4-196z^3+1010z^2-2016z+1764$, and the reduced stationary polynomial $2a'b-ab'$, after excluding inactive $a=0$, is a constant times $H(z)=7z^4-309z^3+2541z^2-8323z+8820$.

Proof of Theorem 3: H is irreducible modulo 13, hence irreducible over the rationals, and the reduction modulo 13 gives a 4-cycle in its Galois group; modulo 19, H factors into irreducible factors of degrees one and three, which gives a 3-cycle. A transitive subgroup of S_4 that contains a 4-cycle and a 3-cycle is S_4. A Sturm sequence shows that exactly one real root of H satisfies $\mu_u(z)a(z)<0$, that is, has $\Gamma^*>0$; the other positive root is inactive. At $z=2$ the reduction is $\Delta(2)=a(2)^2/b(2)=225/107>2$, while the limiting reductions at the roots 1, 4, 9 of μ_u are the squared RMS values of the corresponding components of $\boldsymbol{i}_q$, namely 1, 1 and $1/9$, and the reductions of the one-element limits are zero because the single capacitor and inductor are not useful for this load; hence the global optimum is interior and is the active root, $z=1.9867022429$, with $L=1.0220932870$ and $C=0.4924664876$ in normalized units and $\Delta=2.1029962$. A formula that combines the inner products of the waveforms by sums, products and quotients would return rational values of L and C for these rational data, hence a rational root of H, which does not exist. The normal closure of a tower of quadratic extensions has degree a power of two. If one root of the irreducible H belonged to such a tower, its normal closure

would contain all conjugates and hence the splitting field of H; its degree 24 cannot divide a power of two. Both conclusions use the irreducibility and the group, not the degree alone: the quartic is solvable by radicals (Ferrari), and general radicals are not excluded by this example. ∎

B. Theorem 4 ($m = 4$)

The voltage $u(\tau) = \sqrt{2}\,(8\cos^4\tau + 4\cos^3\tau - 6\cos^2\tau - 2\cos\tau)$ has zero mean and reactive dimension four, with $\mu_u(\xi) = \xi^4 - 30\xi^3 + 273\xi^2 - 820\xi + 576$, and $h(\xi) = (103\xi^3 - 2882\xi^2 + 22327\xi - 39708)/2880$. The reduced stationary polynomial $2a'b - ab'$ is a constant times

$$\begin{aligned} f(z) = {} & 35z^7 + 1003z^6 - 60975z^5 + 1041441z^4 \\ & - 8601960z^3 + 36926304z^2 \\ & - 80912600z + 68173952. \end{aligned} \tag{S6}$$

Proof of Theorem 4: f is irreducible modulo 31, hence irreducible over the rationals, and the reduction modulo 31 gives a 7-cycle in the Galois group. Modulo 29, f factors into irreducible factors of degrees two and five, which gives an element of cycle type $(2)(5)$, whose fifth power is a transposition. A transitive permutation group of prime degree that contains a transposition is the full symmetric group, so the group is S_7, of order 5040, which is not solvable. At $z = 2$ the reduction is $\Delta(2) = 49729/444 \approx 112.0023$, larger than every limiting reduction, the largest of which, at the roots 1 and 4 of μ_u, is 49; hence the global optimum is interior. Exact Sturm root counts give two positive stationary roots: the one in $(1, 4)$ is active, whereas the one near 9.36888 is inactive. Thus the optimum is unique and is a root of f, $z = 2.0024245103$, with $L = 0.1421457641$, $C = 3.5132570384$ in normalized units and $\Delta = 112.0025905$. Since the optimum is a root of an irreducible polynomial whose Galois group is not solvable, no formula by radicals in the inner products of the waveforms gives it, and therefore no such formula gives the optimal single branch for every load. ∎

S-V. Proof of Theorem 5 (Synthesis)

Let $\boldsymbol{q} = \mathsf{D}h(\mathsf{A})\boldsymbol{u}$ with $\deg h < m$, $r(\xi) = \xi\mu_u'(\xi)/m$, $n = \operatorname{rem}(rh, \mu_u)$ and $F_\alpha = (n + \alpha\mu_u)/r$.

(a) The admittance draws $\boldsymbol{q}$. The roots of r are 0 and the $m - 1$ roots of μ_u'. By Rolle's theorem the latter lie strictly between consecutive roots of μ_u, which are positive and simple (Lemma 1), so no root of r is a root of μ_u and $r(\mathsf{A})$ is invertible on $\mathcal{E}_u$, where A has the roots of μ_u as eigenvalues. Since $rh - n$ is a multiple of μ_u and $\mu_u(\mathsf{A})\boldsymbol{u} = \boldsymbol{0}$, $n(\mathsf{A})\boldsymbol{u} = r(\mathsf{A})h(\mathsf{A})\boldsymbol{u}$, hence $F_\alpha(\mathsf{A})\boldsymbol{u} = r(\mathsf{A})^{-1}[n(\mathsf{A}) + \alpha\mu_u(\mathsf{A})]\boldsymbol{u} = h(\mathsf{A})\boldsymbol{u}$ and $\mathsf{Y}_\alpha\boldsymbol{u} = \mathsf{D}h(\mathsf{A})\boldsymbol{u} = \boldsymbol{q}$ for every α.

(b) Positivity. Let $\mu_1, \dots, \mu_{m-1}$ be the roots of μ_u' and $\mu_0 = 0$, all simple and distinct from the roots of μ_u. The Bezoutian form $\operatorname{Bez}(r, b)$ evaluated at two roots of r gives $[r(\mu_j)b(\mu_k) - r(\mu_k)b(\mu_j)]/(\mu_j - \mu_k) = 0$ for $j \neq k$ and, in the limit, $r'(\mu_j)b(\mu_j)$ for $j = k$; in the coordinates $\boldsymbol{v}(\mu_j)$, which form a basis because the μ_j are distinct, both B_n and H_0 are therefore diagonal, with entries $r'(\mu_j)n(\mu_j)$ and $-r'(\mu_j)\mu_u(\mu_j)$. The partial-fraction expansion of F_α, whose denominator r has simple roots, is

$$F_\alpha(\xi) = \alpha + \sum_{j=0}^{m-1} \frac{\rho_j}{\xi - \mu_j}, \quad \rho_j = \frac{n(\mu_j) + \alpha\,\mu_u(\mu_j)}{r'(\mu_j)}, \tag{S7}$$

the constant term being α because $\deg n < \deg r = m$ and $\deg \mu_u = \deg r$ with equal leading coefficients. Hence $B_n - \alpha H_0 \preceq 0$ is equivalent to $\rho_j \leq 0$ for all j. For $H_0 \succ 0$ one needs $\mu_u(\mu_j)/r'(\mu_j) < 0$: at $\mu_0 = 0$, $\mu_u(0)$ has the sign $(-1)^m$ and $r'(0) = \mu_u'(0)/m$ has the sign $(-1)^{m-1}$; at a root μ of μ_u', $r'(\mu) = \mu\mu_u''(\mu)/m$ and $\mu_u''(\mu)/\mu_u(\mu) = -\sum_k(\mu - \lambda_k)^{-2} < 0$, where λ_k are the roots of μ_u. With s as a formal variable for the circuit, $Y_\alpha(s) = sF_\alpha(-s^2) = \alpha s + \gamma_0/s + \sum_{j\geq 1}\gamma_j s/(s^2 + \mu_j)$ with $\gamma_j = -\rho_j \geq 0$ and $\mu_j > 0$; each term is positive real in $\operatorname{Re} s > 0$, so Y_α is a reactance function realized by the Foster network with $C_\infty = \alpha$, $L_0 = 1/\gamma_0$, $L_j = 1/\gamma_j$ and $C_j = \gamma_j/\mu_j$; terms with zero coefficient are omitted.

(c) Gershgorin. With $H_0 = RR^{\mathsf{T}}$ and $M = R^{-1}B_nR^{-\mathsf{T}}$, the condition $B_n - \alpha H_0 \preceq 0$ is $M - \alpha I \preceq 0$, which holds whenever α is at least the largest Gershgorin bound $M_{jj} + \sum_{k\neq j}|M_{jk}|$ of the eigenvalues of M; $\alpha \geq 0$ is enforced by the maximum with zero. The least admissible value in this family is $\max(0, \lambda_{\max}(M))$, which requires an eigenvalue; α_G does not. A Cauer ladder with the same admittance is obtained from the polynomials $n + \alpha\mu_u$ and r by repeated polynomial division, alternately extracting a capacitor and an inductor, without computing the μ_j.

S-VI. The Family for $m = 2$, the Cone and the Incremental Formulas

A. Proposition 4

For $m = 2$, $\mu_u(\xi) = \xi^2 - s\xi + p$ and Proposition 3 gives $\boldsymbol{W}(z) = \pi_2(z\,\mathsf{D}\boldsymbol{u} + \mathsf{AD}\boldsymbol{u}) + \pi_1\mathsf{D}\boldsymbol{u} = (z - s)\mathsf{D}\boldsymbol{u} + \mathsf{AD}\boldsymbol{u}$; from the closure, $\mathsf{AD}\boldsymbol{u} = -\mathsf{D}^3\boldsymbol{u} = s\,\mathsf{D}\boldsymbol{u} + p\,\mathsf{J}\boldsymbol{u}$, so $\boldsymbol{W}(z) = z\,\mathsf{D}\boldsymbol{u} + p\,\mathsf{J}\boldsymbol{u}$ and $\boldsymbol{w}_z = (z\,\mathsf{D}\boldsymbol{u} + p\,\mathsf{J}\boldsymbol{u})/\mu_u(z)$. A capacitor C_p in parallel with a series branch of scale $\gamma_1 = 1/L$ and tuning $\mu_1 = 1/(LC)$ draws $C_p\mathsf{D}\boldsymbol{u} + \gamma_1\boldsymbol{w}_{\mu_1}$. Equating to $\boldsymbol{q}^* = C_\parallel\mathsf{D}\boldsymbol{u} + \Gamma_\parallel\mathsf{J}\boldsymbol{u}$ in the frame $\{\mathsf{D}\boldsymbol{u}, \mathsf{J}\boldsymbol{u}\}$ gives $\gamma_1 p/\mu_u(\mu_1) = \Gamma_\parallel$ and $C_p + \gamma_1\mu_1/\mu_u(\mu_1) = C_\parallel$, that is, $\gamma_1 = \Gamma_\parallel\mu_u(\mu_1)/p$, $L = p/(\Gamma_\parallel\mu_u(\mu_1))$, $C = \gamma_1/\mu_1$ and $C_p = C_\parallel - \Gamma_\parallel\mu_1/p$. With $p > 0$ and $\mu_u(\mu_1) < 0$, L and C are positive if and only if $\Gamma_\parallel < 0$; then $C_p = C_\parallel + |\Gamma_\parallel|\mu_1/p$ is non-negative for every μ_1 when $C_\parallel \geq 0$ and for $\mu_1 \geq pC_\parallel/\Gamma_\parallel$ when $C_\parallel < 0$. At $C_p = 0$, $\mu_1 = pC_\parallel/\Gamma_\parallel$ and the branch is $L = \Gamma_\parallel/K$, $C = K/(pC_\parallel)$ of Theorem 2, since $\mu_u(pC_\parallel/\Gamma_\parallel) = pK/\Gamma_\parallel^2$. The dual family, a shunt inductor $\Gamma_p\mathsf{J}\boldsymbol{u}$ with a series branch, gives $\gamma_1 = C_\parallel\mu_u(\mu_1)/\mu_1$ and $\Gamma_p = \Gamma_\parallel - pC_\parallel/\mu_1$. The series branch is positive for $C_\parallel < 0$ and an intermediate pole; the shunt coefficient Γ_p must also be non-negative, as it is when $\Gamma_\parallel > 0$. For the load of Jeltsema et al., $C_\parallel > 0 > \Gamma_\parallel$, $p = 25$, the roots are 1 and 25, and the published values $L = 0.922$, $C = 0.252$ give $\mu_1 = 4.30$.

B. Proposition 5 and the positive null vector

The set $\mathcal{C}$ is a closed convex cone of the finite-dimensional space $\mathcal{Q}_u$, so the projection of $-\boldsymbol{i}$ onto it exists and is

unique, and its Karush–Kuhn–Tucker conditions are the stated ones: on the active face, the residual is orthogonal to the active generators, which is the projection formula with the blade $\boldsymbol{B}_S$ and the amplitudes $a_j = -\boldsymbol{i} \cdot \boldsymbol{v}^j$ given by the reciprocal frame; on the inactive generators the residual has non-negative inner product. If the generators $\boldsymbol{v}_1, \dots, \boldsymbol{v}_s$ span $\mathcal{Q}_u$ and there is a vector $\boldsymbol{z}_0$ with positive entries such that $\sum_j z_{0,j} \boldsymbol{v}_j = \boldsymbol{0}$ (a positive null vector of the dictionary), then from any amplitudes $\boldsymbol{a}_0$ that draw $\boldsymbol{q}^*$, for instance a least-squares solution, the amplitudes $\boldsymbol{a}_* = \boldsymbol{a}_0 + \beta \boldsymbol{z}_0$ with $\beta = \max_j(-a_{0,j}/z_{0,j})$ are non-negative and draw the same current, with at least one zero amplitude; this is the mechanism that makes the linear program of the fixed-pole design feasible, and it is not available when the dictionary has no positive null vector, in which case the problem is the non-negative least squares of Proposition 5.

C. Adding a branch and readjusting amplitudes

With the bank in the active face S, let $\boldsymbol{r} = (1 - \mathrm{proj}_{\boldsymbol{B}_S})\boldsymbol{i}$ and $\boldsymbol{W}_\perp(z) = (1 - \mathrm{proj}_{\boldsymbol{B}_S})\boldsymbol{W}(z)$. After the amplitudes of the existing branches are re-optimized for every scale of the new branch, the problem in the new branch alone is $\min_{\Gamma \geq 0} \|\boldsymbol{r} + \Gamma \boldsymbol{W}_\perp(z)/\mu_u(z)\|^2$, which has the form of Proposition 3 with $a = \boldsymbol{r} \cdot \boldsymbol{W}_\perp$ and $b = \boldsymbol{W}_\perp^2$, provided $b > 0$. The amplitudes of the existing branches become $a_j(z) = a_j^0 - [\Gamma^*(z)/\mu_u(z)]\, \boldsymbol{W}(z) \cdot \boldsymbol{v}^j$, which equals $a_j^0 + [a(z)/b(z)]\, \boldsymbol{W}(z) \cdot \boldsymbol{v}^j$ only when the new branch is active with $a_j^0 = -\boldsymbol{i} \cdot \boldsymbol{v}^j$; their positivity and the inequalities of Proposition 5 for the inactive generators must be checked, a vanishing amplitude moves the solution to another face, and $b = 0$ means that the new shape is dependent on the active ones. Considering all active faces, including previously inactive branches, gives an exact characterization of adding one branch of free tuning while readjusting amplitudes with the previous tunings fixed. It does not give the best bank with all tunings free, which is not convex.

S-VII. Full Comparison Tables

The tables below list, for the eight cases of Section V of the paper and the thirteen methods, the RMS source current i_s, the number of elements n, the mean stored energy E, the periodic peak element voltage V_{pk} and the periodic peak branch current I_{pk}, together with the status of each design: realizable, certified finite minimum, local search, or an approximation to an unattained infimum at a resonant boundary, which is not a design. The published networks are evaluated with their printed component values. Currents are forced periodic responses at the fixed terminal voltage; in the cases with source impedance the circuit is evaluated coupled, and the fixed-voltage formulas for the capacitor and the parallel pair are marked as applied outside their hypothesis.

S-VIII. Simulation Tables

The tables below give the complete results of the simulation study of Section VI of the paper: the closed-loop runs with three sources, the supervisor with the network in its predictor and admission limits, the fixed banks with losses under both waveforms, the three connection criteria, the estimation of fixed-bank amplitudes from noisy records, and the measured evaluation times. The dictionaries use detuning factors $p = \omega_1^2 LC = 0.14$ and 0.07 (resonant frequencies $\omega_1/\sqrt{p}$), and are fitted once to the waveform indicated and then kept fixed under both waveforms; a zero number of elements means that the positive optimum omits the branch.

The dynamic reference bank has normalized values $C_p = 1/26$, $L_s = 169/175$ and $C_s = 175/351$. Its current is $[1/26 + (175/169)(27/13 - \mathsf{A})^{-1}]\mathsf{D}\boldsymbol{u} = \boldsymbol{q}^*$ for the project $m = 3$ load. With $\omega_1 = 2\pi 50$ rad/s and impedance scale $k_Z = 42.32\ \Omega$, the SI conversion is $L = k_Z L_\tau/\omega_1$ and $C = C_\tau/(k_Z \omega_1)$. The half-admittance branch has twice L_s and half C_s. The dynamic load multipliers are 0.45, 1.35, 0.55 and 1.15, starting at 0, 0.2, 0.4 and 0.6 s with 60 ms raised-cosine transitions. The record has 63 samples per 20 ms period.

In the fixed-bank startup table, $V_{C,\mathrm{start}}$ covers only series LC branches, connected from zero state at their minimum-mismatch phase and observed for five periods; simple branches are already periodic. A reported zero for a capacitor-only bank therefore means no series-branch startup was evaluated. In the separate manoeuvre table, cases 1/2/3 and 4 start with zero capacitor voltage, whereas case 5 starts at 6.364 V. Cases 1/2/3 share the same physical conditions and are grouped once. Waiting times and slopes are maxima over the requested manoeuvres.

TABLE SI
EXAMPLE OF SECTION III-C ($m = 2$, PHYSICAL COMPONENTS AT 50 HZ): ALL METHODS

Method	i_s (A)	n	E (J)	V_{pk} (V)	I_{pk} (A)	Status
No compensator	0.5503	0	–	–	–	realizable
Capacitor (Kusters–Moore)	0.5049	1	0.000205	1.01	0.366	realizable
Parallel pair (Page), cone	0.5049	1	0.000205	1.01	0.366	realizable
Series branch, Theorem 2	0.4932	2	0.000896	2.17	0.364	realizable
Series branch, direct search	0.4932	2	0.000896	2.17	0.364	local search, not certified
Series branch, energy-bounded search	0.4963	2	0.000488	1.7	0.333	energy-bounded local search
Series branch, L fixed (best of a set)	0.4932	2	0.000871	2.12	0.365	best of fixed L values
Theorem 5, Cauer ladder	0.4932	4	0.000488	1.61	0.301	realizable
Theorem 5, Foster form	0.4932	4	0.000488	1.68	0.29	realizable
Foster, prescribed 20 % frequency offsets	0.4932	4	0.00119	2.89	0.299	realizable
Poles 5 % from the frequencies (Jeon)	0.4932	4	0.00466	9.28	0.331	realizable
Fixed intermediate poles, linear program	0.4932	3	0.000538	1.73	0.307	realizable
Capacitor optimized in the coupled circuit	0.5049	1	0.000205	1.01	0.366	realizable

TABLE SII
LOAD OF THEOREM 3 WITH $G = 0.8$ ($m = 3$, PHYSICAL COMPONENTS AT 50 HZ): ALL METHODS

Method	i_s (A)	n	E (J)	V_{pk} (V)	I_{pk} (A)	Status
No compensator	2.008	0	–	–	–	realizable
Capacitor (Kusters–Moore)	2.008	0	–	–	–	realizable
Parallel pair (Page), cone	2.008	0	–	–	–	realizable
Series branch, Theorem 2	–	–	–	–	–	not applicable
Series branch, direct search	1.389	2	0.00742	4.22	2.5	local search, not certified
Series branch, energy-bounded search	1.389	2	0.00742	4.22	2.5	energy-bounded local search
Series branch, L fixed (best of a set)	1.412	2	0.00851	4.26	2.91	best of fixed L values
Theorem 5, Cauer ladder	1.386	6	0.0103	8.18	2.87	realizable
Theorem 5, Foster form	1.386	6	0.0103	8.01	2.79	realizable
Foster, prescribed 20 % frequency offsets	1.386	6	0.0223	9.02	1.34	realizable
Poles 5 % from the frequencies (Jeon)	1.386	6	0.0475	16.3	1.41	realizable
Fixed intermediate poles, linear program	1.386	5	0.00886	6.87	2.7	realizable
Capacitor optimized in the coupled circuit	2.008	0	–	–	–	realizable

TABLE SIII
230 V WITH VOLTAGE THD 4.24 %, 1000 W ($m = 3$): ALL METHODS

Method	i_s (A)	n	E (J)	V_{pk} (V)	I_{pk} (A)	Status
No compensator	6.958	0	–	–	–	realizable
Capacitor (Kusters–Moore)	4.408	1	1.96	344	7.07	realizable
Parallel pair (Page), cone	4.408	1	1.96	344	7.07	realizable
Series branch, Theorem 2	–	–	–	–	–	not applicable
Series branch, direct search	4.348	2	6.04	663	7.79	local search, not certified
Series branch, energy-bounded search	4.35	2	4.83	570	7.82	energy-bounded local search
Series branch, L fixed (best of a set)	4.348	2	8.19	837	7.74	best of fixed L values
Theorem 5, Cauer ladder	4.348	6	9.64	364	14.7	realizable
Theorem 5, Foster form	4.348	6	9.64	579	10.1	realizable
Foster, prescribed 20 % frequency offsets	4.348	6	18.1	1.88e+03	6.46	realizable
Poles 5 % from the frequencies (Jeon)	4.348	6	37.8	3.5e+03	7.07	realizable
Fixed intermediate poles, linear program	4.348	5	4.83	594	6.97	realizable
Capacitor optimized in the coupled circuit	4.408	1	1.96	344	7.07	realizable

TABLE SIV
R–L LOAD OF JELTSEMA ET AL. ($m = 2$, $\omega_1 = 1$ RAD/S): ALL METHODS

Method	i_s (A)	n	E (J)	V_{pk} (V)	I_{pk} (A)	Status
No compensator	4.5	0	–	–	–	realizable
Capacitor (Kusters–Moore)	4.058	1	4.52	21.2	3.58	realizable
Parallel pair (Page), cone	4.058	1	4.52	21.2	3.58	realizable
Series branch, Theorem 2	–	–	–	–	–	not applicable
Series branch, direct search	2.061	2	3.98e+05	1.41e+05	5.66	resonant infimum, not attained
Series branch, energy-bounded search	2.585	2	28.4	19.6	4.49	energy-bounded local search
Series branch, L fixed (best of a set)	2.071	2	418	154	5.6	best of fixed L values
Theorem 5, Cauer ladder	2.001	4	34.2	21.2	10.9	realizable
Theorem 5, Foster form	2.001	4	34.2	21.2	10.9	realizable
Foster, prescribed 20 % frequency offsets	2.001	4	111	45.9	5.29	realizable
Poles 5 % from the frequencies (Jeon)	2.001	4	414	152	5.57	realizable
Fixed intermediate poles, linear program	2.001	3	28.4	21.2	7.4	realizable
Capacitor optimized in the coupled circuit	4.058	1	4.52	21.2	3.58	realizable
Published Jeltsema 2014	2.001	3	31.8	21.2	5.68	published components

TABLE SV
LOAD OF CZARNECKI 1981, EXACT FIG. 1 ($m = 3$): ALL METHODS

Method	i_s (A)	n	E (J)	V_{pk} (V)	I_{pk} (A)	Status
No compensator	12.24	0	–	–	–	realizable
Capacitor (Kusters–Moore)	12.24	0	–	–	–	realizable
Parallel pair (Page), cone	12.24	0	–	–	–	realizable
Series branch, Theorem 2	–	–	–	–	–	not applicable
Series branch, direct search	7.502	2	1.27e+03	331	16.5	certified finite global minimum
Series branch, energy-bounded search	7.502	2	1.27e+03	331	16.5	energy-bounded local search
Series branch, L fixed (best of a set)	7.538	2	1.34e+03	331	17.7	best of fixed L values
Theorem 5, Cauer ladder	3.097	6	2.81e+03	264	34.3	realizable
Theorem 5, Foster form	3.097	6	2.81e+03	332	23.7	realizable
Foster, prescribed 20 % frequency offsets	3.097	6	3.22e+03	839	14.1	realizable
Poles 5 % from the frequencies (Jeon)	3.097	6	8.06e+03	1.54e+03	12.3	realizable
Fixed intermediate poles, linear program	3.097	5	3.3e+03	331	27.4	realizable
Capacitor optimized in the coupled circuit	12.24	0	–	–	–	realizable
Published Czarnecki 1981 Fig3	3.097	4	1.82e+03	637	23.5	published components
Published Czarnecki 1981 eq. (13)	3.097	4	1.81e+03	497	15.6	published components

TABLE SVI
LOAD OF CZARNECKI 1981, ADMITTANCES OF HIS EQ. (12): ALL METHODS

Method	i_s (A)	n	E (J)	V_{pk} (V)	I_{pk} (A)	Status
No compensator	12.24	0	–	–	–	realizable
Capacitor (Kusters–Moore)	12.24	0	–	–	–	realizable
Parallel pair (Page), cone	12.24	0	–	–	–	realizable
Series branch, Theorem 2	–	–	–	–	–	not applicable
Series branch, direct search	7.501	2	1.27e+03	331	16.5	certified finite global minimum
Series branch, energy-bounded search	7.501	2	1.27e+03	331	16.5	energy-bounded local search
Series branch, L fixed (best of a set)	7.536	2	1.34e+03	331	17.7	best of fixed L values
Theorem 5, Cauer ladder	3.094	6	2.81e+03	264	34.3	realizable
Theorem 5, Foster form	3.094	6	2.81e+03	332	23.7	realizable
Foster, prescribed 20 % frequency offsets	3.094	6	3.22e+03	839	14.1	realizable
Poles 5 % from the frequencies (Jeon)	3.094	6	8.06e+03	1.54e+03	12.3	realizable
Fixed intermediate poles, linear program	3.094	5	3.3e+03	331	27.4	realizable
Capacitor optimized in the coupled circuit	12.24	0	–	–	–	realizable

TABLE SVII
LOAD OF CZARNECKI 1987, IDEAL SOURCE ($m = 3$): ALL METHODS

Method	i_s (A)	n	E (J)	V_{pk} (V)	I_{pk} (A)	Status
No compensator	100.0	0	–	–	–	realizable
Capacitor (Kusters–Moore)	53.19	1	4.14e+03	149	123	realizable
Parallel pair (Page), cone	53.19	1	4.14e+03	149	123	realizable
Series branch, Theorem 2	–	–	–	–	–	not applicable
Series branch, direct search	50.01	2	1.36e+07	2.22e+05	122	resonant infimum, not attained
Series branch, energy-bounded search	51.11	2	5e+03	155	106	energy-bounded local search
Series branch, L fixed (best of a set)	50.35	2	5.84e+03	164	118	best of fixed L values
Theorem 5, Cauer ladder	50	6	5.16e+03	150	76.1	realizable
Theorem 5, Foster form	50	6	5.16e+03	151	71	realizable
Foster, prescribed 20 % frequency offsets	50	6	5e+03	156	80.5	realizable
Poles 5 % from the frequencies (Jeon)	50	6	8.81e+04	1.52e+03	121	realizable
Fixed intermediate poles, linear program	50	5	5.1e+03	156	79.4	realizable
Capacitor optimized in the coupled circuit	53.19	1	4.14e+03	149	123	realizable

TABLE SVIII
LOAD OF CZARNECKI 1987 WITH THE SOURCE IMPEDANCE OF HIS EXAMPLE: ALL METHODS

Method	i_s (A)	n	E (J)	V_{pk} (V)	I_{pk} (A)	Status
No compensator	95.47	0	–	–	–	realizable
Capacitor (Kusters–Moore)	227.5	1	5.41e+03	218	454	outside hypothesis (H1)
Parallel pair (Page), cone	227.5	1	5.41e+03	218	454	outside hypothesis (H1)
Series branch, Theorem 2	–	–	–	–	–	not applicable
Series branch, direct search	49.73	2	1.3e+07	2.17e+05	120	boundary approximation; globality unproved
Series branch, energy-bounded search	50.19	2	4.95e+03	153	111	energy-bounded local search
Series branch, L fixed (best of a set)	49.91	2	5.67e+03	163	117	best of fixed L values
Theorem 5, Cauer ladder	49.74	6	5.11e+03	149	75.6	realizable
Theorem 5, Foster form	49.74	6	5.11e+03	150	71.5	realizable
Foster, prescribed 20 % frequency offsets	49.74	6	4.95e+03	155	81	realizable
Poles 5 % from the frequencies (Jeon)	49.74	6	8.72e+04	1.51e+03	121	realizable
Fixed intermediate poles, linear program	49.74	5	5.04e+03	155	79.1	realizable
Capacitor optimized in the coupled circuit	66.55	1	2.9e+03	147	158	coupled-circuit numerical search
Published 1987 no bank	95.47	0	–	–	–	published components
Published 1987 C=.61	66.77	1	2.99e+03	149	162	published components
Published 1987 L=.06	51.39	2	5.02e+03	145	123	published components
Published 1987 L=.10	50.34	2	5.07e+03	150	116	published components
Published 1987 L=.20	49.92	2	5.77e+03	163	119	published components
Published 1987 Foster eq. (18)	49.73	6	4.82e+03	156	78.3	published components

TABLE SIX
CLOSED-LOOP RUNS OF 0.8 S WITH THE MODERATE WAVEFORM: RMS SOURCE CURRENT OVER THE RUN, PEAK BRANCH CURRENT, PEAK CAPACITOR VOLTAGE, BANK LOSSES AND MANOEUVRES, FOR THE TWO SUPERVISORS AND THE FOUR FIXED STATES, WITH THREE SOURCES

Source	Strategy	i_s (A)	I_{pk} (A)	$V_{C,\mathrm{pk}}$ (V)	P_{bank} (W)	Man.
ideal	Projection	5.238	14.36	1017	24.65	9
ideal	Fundamental	5.238	14.36	1017	24.65	9
ideal	fixed 00	6.274	0.00	0	0.10	0
ideal	fixed 10	5.049	7.18	1017	12.11	1
ideal	fixed 01	5.298	14.36	1017	24.12	1
ideal	fixed 11	6.862	14.36	1017	36.13	2
0.4 Ω, 0.8 mH	Projection	5.090	13.95	1008	23.53	9
0.4 Ω, 0.8 mH	Fundamental	5.090	13.95	1008	23.53	9
0.4 Ω, 0.8 mH	fixed 00	6.276	0.00	0	0.09	0
0.4 Ω, 0.8 mH	fixed 10	5.029	7.10	1008	11.43	1
0.4 Ω, 0.8 mH	fixed 01	5.094	14.10	1005	22.25	1
0.4 Ω, 0.8 mH	fixed 11	6.385	14.02	1001	32.83	2
1.5 Ω, 3 mH	Projection	4.830	13.01	1006	21.03	9
1.5 Ω, 3 mH	Fundamental	4.830	13.01	1006	21.03	9
1.5 Ω, 3 mH	fixed 00	6.280	0.00	0	0.09	0
1.5 Ω, 3 mH	fixed 10	5.011	6.94	1006	10.40	1
1.5 Ω, 3 mH	fixed 01	4.949	13.62	994	20.38	1
1.5 Ω, 3 mH	fixed 11	6.165	13.35	983	30.59	2

TABLE SX
SUPERVISOR WITH THE NETWORK IN ITS PREDICTOR AND ADMISSION LIMITS (MODERATE WAVEFORM, 0.8 S)

Source	Control	V limit (V)	I limit (A)	i_s (A)	I_{pk} (A)	Man.
0.4 Ω, 0.8 mH	terminal cost, previous policy	–	–	5.090	13.95	9
0.4 Ω, 0.8 mH	terminal cost	–	–	5.103	14.02	3
0.4 Ω, 0.8 mH	network cost	–	–	5.103	14.02	3
0.4 Ω, 0.8 mH	network cost, limits	1100	16	5.029	7.10	1
0.4 Ω, 0.8 mH	network cost, limits	1100	8	5.029	7.10	1
0.4 Ω, 0.8 mH	network cost, limits	900	16	6.276	0.00	0
1.5 Ω, 3 mH	terminal cost, previous policy	–	–	4.830	13.01	9
1.5 Ω, 3 mH	terminal cost	–	–	4.893	13.00	3
1.5 Ω, 3 mH	network cost	–	–	4.893	13.00	3
1.5 Ω, 3 mH	network cost, limits	1100	16	5.019	6.88	1
1.5 Ω, 3 mH	network cost, limits	1100	8	5.019	6.88	1
1.5 Ω, 3 mH	network cost, limits	900	16	6.280	0.00	0

TABLE SXI
FIXED BANKS AT 230 V AND 1000 W WITH LOSSES ($Q_L = 100$, $Q_C = 500$ AT 50 HZ): STEADY-STATE SOURCE CURRENT, LOSSES, PERIODIC PEAK ELEMENT VOLTAGE AND STARTUP PEAK CAPACITOR VOLTAGE

Waveform	Design	n	i_s (A)	P_{loss} (W)	V_{pk} (V)	$V_{C,\mathrm{start}}$ (V)
original	No bank	0	6.300	0.00	0	0
moderate	No bank	0	6.958	0.00	0	0
original	Family z=1.45	4	4.504	35.41	676	1072
moderate	Family z=1.45	4	4.720	85.55	1050	1825
original	Family z=1.8	4	4.429	18.53	563	736
moderate	Family z=1.8	4	4.492	33.20	737	1220
original	Family z=2.04115	4	4.408	13.93	571	609
moderate	Family z=2.04115	4	4.424	17.60	644	1046
original	Family three elements	3	4.406	13.46	576	594
moderate	Family three elements	3	4.417	15.88	634	1020
original	half-admittance LC	3	4.980	6.96	576	594
moderate	half-admittance LC	3	5.102	7.99	634	1020
original	Jeon, three LC	6	4.737	88.38	2159	3089
moderate	Jeon, three LC	6	4.938	135.55	3473	4333
original	p=14% fitted	2	6.301	0.66	1118	1143
moderate	p=14% fitted	2	6.944	0.02	363	540
original	p=7% fitted	0	6.300	0.00	0	0
moderate	p=7% fitted	0	6.958	0.00	0	0
original	p=14% fitted to moderate waveform	3	11.714	50.07	1118	1143
moderate	p=14% fitted to moderate waveform	3	4.397	3.14	363	540
original	p=7% fitted to moderate waveform	1	14.591	11.50	563	0
moderate	p=7% fitted to moderate waveform	1	4.419	2.49	344	0

TABLE SXII
UNSCALED BRANCH MANOEUVRES ($\sqrt{3}$ V RMS): THREE CRITERIA UNDER THREE PHYSICAL CONDITIONS (PEAK BRANCH CURRENT AND CAPACITOR VOLTAGE, WAITING TIME AND INITIAL SLOPE)

Case	Criterion	R (Ω)	R_b (Ω)	I_{pk} (A)	$V_{C,\mathrm{pk}}$ (V)	wait (ms)	di/dt (A/s)
1/2/3	immediate	0.05	1000	3.690	5.317	0.0	1.08e+03
1/2/3	valve voltage	0.05	1000	3.931	5.794	1.5	0.000244
1/2/3	mismatch energy	0.05	1000	2.927	4.158	18.9	1.48e+03
4	immediate	0	–	3.963	5.734	0.0	1.08e+03
4	valve voltage	0	–	4.482	6.434	18.2	0.000291
4	mismatch energy	0	–	3.088	4.533	19.0	1.56e+03
5	immediate	0.05	1000	5.223	7.570	0.0	963
5	valve voltage	0.05	1000	5.080	7.242	19.0	639
5	mismatch energy	0.05	1000	5.038	7.130	2.4	2.53e+03

TABLE SXIII
FIXED-BANK AMPLITUDES FROM NOISY RECORDS: RELATIVE AMPLITUDE-COEFFICIENT ERROR AND EXCESS OF SQUARED SOURCE CURRENT OVER THE OPTIMUM, MEAN OF TWO SEEDS

	Relative coefficient error				Excess $i_s^2 - i_*^2$ (A^2)			
Method	0 %	0.1 %	1 %	3 %	0 %	0.1 %	1 %	3 %
one period	1.2e-09	0.0014	0.055	0.22	4e-13	6e-07	0.0009	0.01
average of 8 periods	4.4e-10	0.00015	0.0085	0.062	4e-13	3e-08	2e-05	0.001
regularized fit	4.4e-10	0.00011	0.001	0.003	4e-13	5e-08	3e-06	3e-05
mSOGI, $a = 40$	6.5e-05	0.00021	0.0015	0.0044	9e-09	1e-07	1e-05	9e-05
mSOGI, $a = 100$	3.3e-07	0.0003	0.003	0.0089	3e-13	2e-07	2e-05	0.0002

TABLE SXIV
MEASURED TIME PER EVALUATION OF ALL STATES ON THE SAME HOST (MATLAB R2024A, INTEL I9-9880H; ORIENTATIVE), IN MICROSECONDS

N	M	states	time domain	full FFT	FFT, known support	direct enumeration
63	2	4	1.2	6.2	4.6	1.8
63	4	16	1.1	5.7	3.9	2.9
63	8	256	1.5	5.7	4.9	51.4
255	2	4	1.0	8.5	6.9	2.5
255	4	16	1.2	10.2	7.6	5.3
255	8	256	2.1	11.6	8.7	69.9
1023	2	4	2.7	71.8	38.5	10.6
1023	4	16	4.6	76.1	41.2	32.1
1023	8	256	4.8	56.8	32.3	205.8